g_m

CHAPTER

FET Small-Signal Analysis

9

9.1 INTRODUCTION

Field-effect transistor amplifiers provide an excellent voltage gain with the added feature of a high input impedance. They are also considered low-power consumption configurations with good frequency range and minimal size and weight. Both JFET and depletion MOSFET devices can be used to design amplifiers having similar voltage gains. The depletion MOSFET circuit, however, has a much higher input impedance than a similar JFET configuration.

While a BJT device controls a large output (collector) current by means of a relatively small input (base) current, the FET device controls an output (drain) current by means of a small input (gate-voltage) voltage. In general, therefore, the BJT is a *current-controlled* device and the FET is a *voltage-controlled* device. In both cases, however, note that the output current is the controlled variable. Because of the high input characteristic of FETs, the ac equivalent model is somewhat simpler than that employed for BJTs. While the BJT had an amplification factor β (beta), the FET has a transconductance factor, g_m.

The FET can be used as a linear amplifier or as a digital device in logic circuits. In fact, the enhancement MOSFET is quite popular in digital circuitry, especially in CMOS circuits that require very low power consumption. FET devices are also widely used in high-frequency applications and in buffering (interfacing) applications. Table 9.1, located at the end of the chapter, provides a summary of FET small-signal amplifier circuits and related formulas.

While the common-source configuration is the most popular providing an inverted, amplified signal, one also finds common-drain (source-follower) circuits providing unity gain with no inversion and common-gate circuits providing gain with no inversion. As with BJT amplifiers, the important circuit features described in this chapter include voltage gain, input impedance, and output impedance. Due to the very high input impedance, the input current is generally assumed to be 0 μA and the current gain is an undefined quantity. While the voltage gain of an FET amplifier is generally less than that obtained using a BJT amplifier, the FET amplifier provides a much higher input impedance than that of a BJT configuration. Output impedance values are comparable for both BJT and FET circuits.

FET ac amplifier networks can also be analyzed using computer software. Using PSpice, one can perform a dc analysis to obtain the circuit bias conditions and an ac analysis to determine the small-signal voltage gain. Using PSpice transistor models,

one can analyze the circuit using specific transistor models. On the other hand, one can develop a program using a language such as BASIC that can perform both the dc and ac analyses and provide the results in a very special format.

9.2 FET SMALL-SIGNAL MODEL

The ac analysis of an FET configuration requires that a small-signal ac model for the FET be developed. A major component of the ac model will reflect the fact that an ac voltage applied to the input gate-to-source terminals will control the level of current from drain to source.

The gate-to-source voltage controls the drain-to-source (channel) current of an FET.

Recall from Chapter 6 that a dc gate-to-source voltage controlled the level of dc drain current through a relationship known as Shockley's equation: $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$. The *change* in collector current that will result from a *change* in gate-to-source voltage can be determined using the transconductance factor g_m in the following manner:

$$\Delta I_D = g_m \, \Delta V_{GS} \tag{9.1}$$

The prefix *trans-* in the terminology applied to g_m reveals that it establishes a relationship between an output and input quantity. The root word *conductance* was chosen because g_m is determined by a voltage-to-current ratio similar to the ratio that defines the conductance of a resistor $G = 1/R = I/V$.

Solving for g_m in Eq. (9.1), we have:

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} \tag{9.2}$$

Graphical Determination of g_m

If we now examine the transfer characteristics of Fig. 9.1, we find that g_m is actually the slope of the characteristics at the point of operation. That is,

$$g_m = m = \frac{\Delta y}{\Delta x} = \frac{\Delta I_D}{\Delta V_{GS}} \tag{9.3}$$

Following the curvature of the transfer characteristics, it is reasonably clear that the slope and, therefore, g_m increase as we progress from V_P to I_{DSS}. Or, in other words, as V_{GS} approaches 0 V, the magnitude of g_m increases.

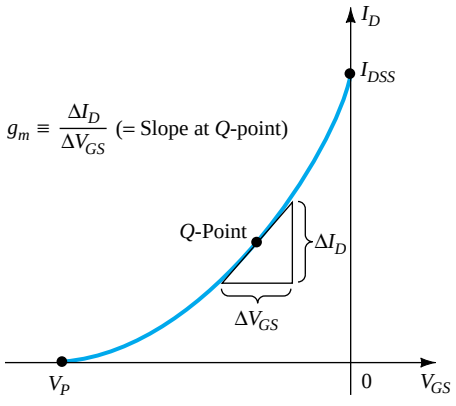


Figure 9.1 Definition of g_m using transfer characteristic.

Equation (9.2) reveals that g_m can be determined at any Q-point on the transfer characteristics by simply choosing a finite increment in V_{GS} (or in I_D) about the Q-point and then finding the corresponding change in I_D (or V_{GS}, respectively). The resulting changes in each quantity are then substituted in Eq. (9.2) to determine g_m.

EXAMPLE 9.1

Determine the magnitude of g_m for a JFET with $I_{DSS} = 8$ mA and $V_P = -4$ V at the following dc bias points:
(a) $V_{GS} = -0.5$ V.
(b) $V_{GS} = -1.5$ V.
(c) $V_{GS} = -2.5$ V.

Solution

The transfer characteristics are generated as Fig. 9.2 using the procedure defined in Chapter 6. Each operating point is then identified and a tangent line is drawn at each point to best reflect the slope of the transfer curve in this region. An appropriate increment is then chosen for V_{GS} to reflect a variation to either side of each Q-point. Equation (9.2) is then applied to determine g_m.

(a) $$g_m = \frac{\Delta I_D}{\Delta V_{GS}} \cong \frac{2.1\text{ mA}}{0.6\text{ V}} = \mathbf{3.5\text{ mS}}$$

(b) $$g_m = \frac{\Delta I_D}{\Delta V_{GS}} \cong \frac{1.8\text{ mA}}{0.7\text{ V}} \cong \mathbf{2.57\text{ mS}}$$

(c) $$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{1.5\text{ mA}}{1.0\text{ V}} = \mathbf{1.5\text{ mS}}$$

Note the decrease in g_m as V_{GS} approaches V_P.

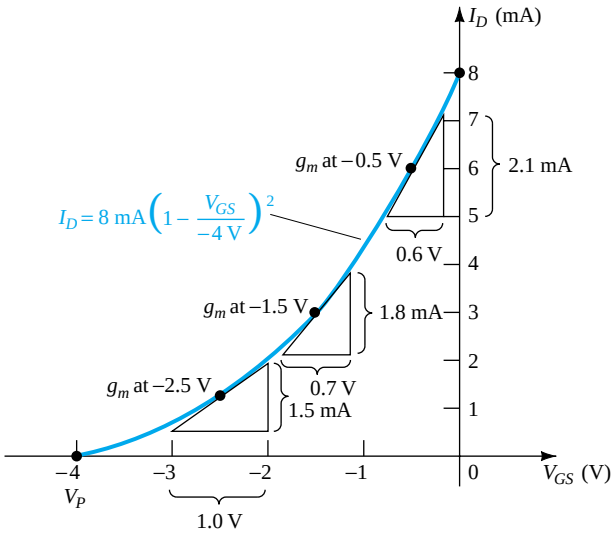


Figure 9.2 Calculating g_m at various bias points.

Mathematical Definition of g_m

The graphical procedure just described is limited by the accuracy of the transfer plot and the care with which the changes in each quantity can be determined. Naturally, the larger the graph the better the accuracy, but this can then become a cumbersome

problem. An alternative approach to determining g_m employs the approach used to find the ac resistance of a diode in Chapter 1, where it was stated that:

The derivative of a function at a point is equal to the slope of the tangent line drawn at that point.

If we therefore take the derivative of I_D with respect to V_{GS} (differential calculus) using Shockley's equation, an equation for g_m can be derived as follows:

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}\bigg|_{Q\text{-pt.}} = \frac{dI_D}{dV_{GS}}\bigg|_{Q\text{-pt.}} = \frac{d}{dV_{GS}}\left[I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2\right]$$

$$= I_{DSS}\frac{d}{dV_{GS}}\left(1 - \frac{V_{GS}}{V_P}\right)^2 = 2I_{DSS}\left[1 - \frac{V_{GS}}{V_P}\right]\frac{d}{dV_{GS}}\left(1 - \frac{V_{GS}}{V_P}\right)$$

$$= 2I_{DSS}\left[1 - \frac{V_{GS}}{V_P}\right]\left[\frac{d}{dV_{GS}}(1) - \frac{1}{V_P}\frac{dV_{GS}}{dV_{GS}}\right] = 2I_{DSS}\left[1 - \frac{V_{GS}}{V_P}\right]\left[0 - \frac{1}{V_P}\right]$$

and

$$\boxed{g_m = \frac{2I_{DSS}}{|V_P|}\left[1 - \frac{V_{GS}}{V_P}\right]} \tag{9.4}$$

where $|V_P|$ denotes magnitude only to ensure a positive value for g_m.

It was mentioned earlier that the slope of the transfer curve is a maximum at $V_{GS} = 0$ V. Plugging in $V_{GS} = 0$ V into Eq. (9.4) will result in the following equation for the maximum value of g_m for a JFET in which I_{DSS} and V_P have been specified:

$$g_m = \frac{2I_{DSS}}{|V_P|}\left[1 - \frac{0}{V_P}\right]$$

and

$$\boxed{g_{m0} = \frac{2I_{DSS}}{|V_P|}} \tag{9.5}$$

where the added subscript 0 reminds us that it is the value of g_m when $V_{GS} = 0$ V. Equation (9.4) then becomes

$$\boxed{g_m = g_{m0}\left[1 - \frac{V_{GS}}{V_P}\right]} \tag{9.6}$$

EXAMPLE 9.2

For the JFET having the transfer characteristics of Example 9.1:
(a) Find the maximum value of g_m.
(b) Find the value of g_m at each operating point of Example 9.1 using Eq. (9.6) and compare with the graphical results.

Solution

(a) $g_{m0} = \dfrac{2I_{DSS}}{|V_P|} = \dfrac{2(8\text{ mA})}{4\text{ V}} = \mathbf{4\ mS}$ (maximum possible value of g_m)

(b) At $V_{GS} = -0.5$ V,

$$g_m = g_{m0}\left[1 - \frac{V_{GS}}{V_P}\right] = 4\text{ mS}\left[1 - \frac{-0.5\text{ V}}{-4\text{ V}}\right] = \mathbf{3.5\ mS}$$ (versus 3.5 mS graphically)

At $V_{GS} = -1.5$ V,

$$g_m = g_{m0}\left[1 - \frac{V_{GS}}{V_P}\right] = 4\text{ mS}\left[1 - \frac{-1.5\text{ V}}{-4\text{ V}}\right] = \mathbf{2.5\ mS} \quad \text{(versus 2.57 mS graphically)}$$

At $V_{GS} = -2.5$ V,

$$g_m = g_{m0}\left[1 - \frac{V_{GS}}{V_P}\right] = 4\text{ mS}\left[1 - \frac{-2.5\text{ V}}{-4\text{ V}}\right] = \mathbf{1.5\ mS} \quad \text{(versus 1.5 mS graphically)}$$

The results of Example 9.2 are certainly sufficiently close to validate Eq. (9.4) through (9.6) for future use when g_m is required.

On specification sheets, g_m is provided as y_{fs} where y indicates it is part of an admittance equivalent circuit. The f signifies *f*orward transfer parameter, and the s reveals that it is connected to the source terminal.

In equation form,

$$\boxed{g_m = y_{fs}} \tag{9.7}$$

For the JFET of Fig. 5.18, y_{fs} ranges from 1000 to 5000 μS or 1 to 5 mS.

Plotting g_m vs. V_{GS}

Since the factor $\left(1 - \frac{V_{GS}}{V_P}\right)$ of Eq. (9.6) is less than 1 for any value of V_{GS} other than 0 V, the magnitude of g_m will decrease as V_{GS} approaches V_P and the ratio $\frac{V_{GS}}{V_P}$ increases in magnitude. At $V_{GS} = V_P$, $g_m = g_{m0}(1 - 1) = 0$. Equation (9.6) defines a straight line with a minimum value of 0 and a maximum value of g_m as shown by the plot of Fig. 9.3.

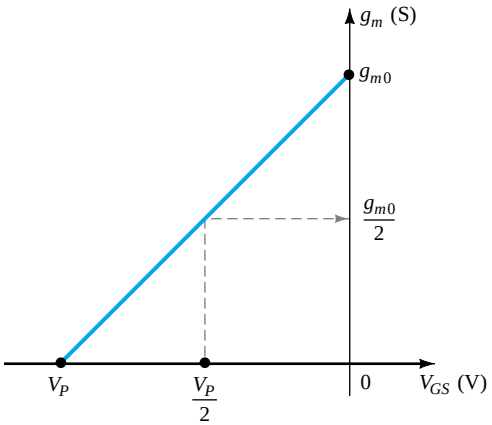


Figure 9.3 Plot of g_m vs. V_{GS}.

Figure 9.3 also reveals that when V_{GS} is one-half the pinch-off value, g_m will be one-half the maximum value.

EXAMPLE 9.3

Plot g_m vs. V_{GS} for the JFET of examples 9.1 and 9.2.

Solution

Note Fig. 9.4.

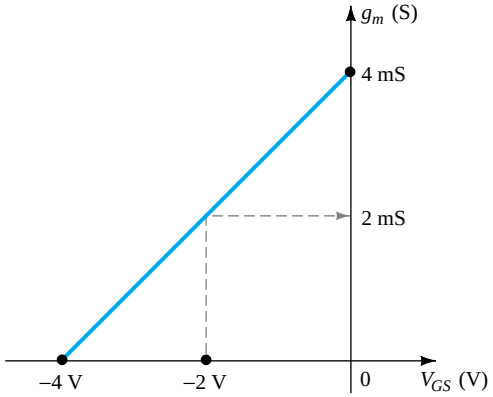


Figure 9.4 Plot of g_m vs. V_{GS} for a JFET with $I_{DSS} = 8$ mA and $V_P = -4$ V.

Impact of I_D on g_m

A mathematical relationship between g_m and the dc bias current I_D can be derived by noting that Shockley's equation can be written in the following form:

$$1 - \frac{V_{GS}}{V_P} = \sqrt{\frac{I_D}{I_{DSS}}} \tag{9.8}$$

Substituting Eq. (9.8) into Eq. (9.6) will result in

$$g_m = g_{m0}\left(1 - \frac{V_{GS}}{V_P}\right) = g_{m0}\sqrt{\frac{I_D}{I_{DSS}}} \tag{9.9}$$

Using Eq. (9.9) to determine g_m for a few specific values of I_D, the results are

(a) If $I_D = I_{DSS}$,

$$g_m = g_{m0}\sqrt{\frac{I_{DSS}}{I_{DSS}}} = \boldsymbol{g_{m0}}$$

(b) If $I_D = I_{DSS}/2$,

$$g_m = g_{m0}\sqrt{\frac{I_{DSS}/2}{I_{DSS}}} = \mathbf{0.707}\boldsymbol{g_{m0}}$$

(c) If $I_D = I_{DSS}/4$,

$$g_m = g_{m0}\sqrt{\frac{I_{DSS}/4}{I_{DSS}}} = \frac{g_{m0}}{2} = \mathbf{0.5}\boldsymbol{g_{m0}}$$

EXAMPLE 9.4 Plot g_m vs. I_D for the JFET of Examples 9.1 through 9.3.

Solution

See Fig. 9.5.

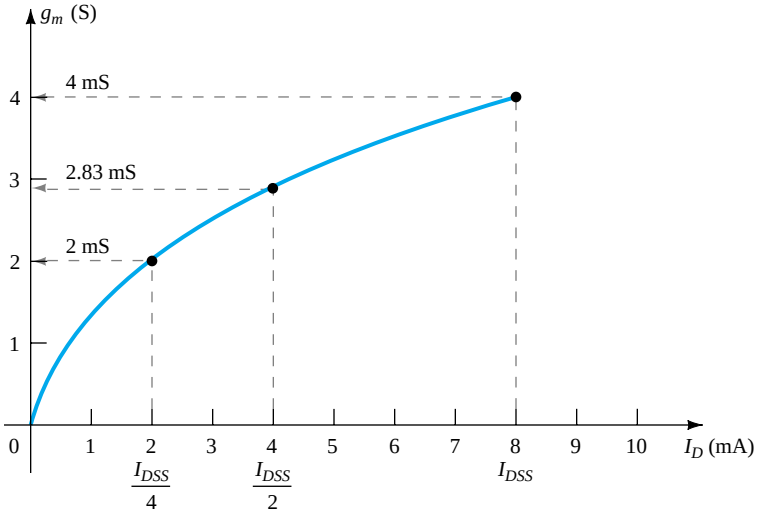


Figure 9.5 Plot of g_m vs. I_D for a JFET with $I_{DSS} = 8$ mA and $V_{GS} = -4$ V.

The plots of Examples 9.3 and 9.4 clearly reveal that the highest values of g_m are obtained when V_{GS} approaches 0 V and I_D its maximum value of I_{DSS}.

FET Input Impedance Z_i

The input impedance of all commercially available FETs is sufficiently large to assume that the input terminals approximate an open circuit. In equation form,

$$Z_i \text{ (FET)} = \infty\ \Omega \tag{9.10}$$

For a JFET a practical value of $10^9\ \Omega$ (1000 MΩ) is typical, while a value of 10^{12} to $10^{15}\ \Omega$ is typical for MOSFETs.

FET Output Impedance Z_o

The output impedance of FETs is similar in magnitude to that of conventional BJTs. On FET specification sheets, the output impedance will typically appear as y_{os} with the units of μS. The parameter y_{os} is a component of an *admittance equivalent circuit*, with the subscript *o* signifying an *o*utput network parameter and *s* the terminal (source) to which it is attached in the model. For the JFET of Fig. 5.18, y_{os} has a range of 10 to 50 μS or 20 kΩ ($R = 1/G = 1/50\ \mu$S) to 100 kΩ ($R = 1/G = 1/10\ \mu$S).

In equation form,

$$Z_o \text{ (FET)} = r_d = \frac{1}{y_{os}} \tag{9.11}$$

The output impedance is defined on the characteristics of Fig. 9.6 as the slope of the horizontal characteristic curve at the point of operation. The more horizontal the curve, the greater the output impedance. If perfectly horizontal, the ideal situation is on hand with the output impedance being infinite (an open circuit)—an often applied approximation.

In equation form,

$$r_d = \left.\frac{\Delta V_{DS}}{\Delta I_D}\right|_{V_{GS}=\text{constant}} \tag{9.12}$$

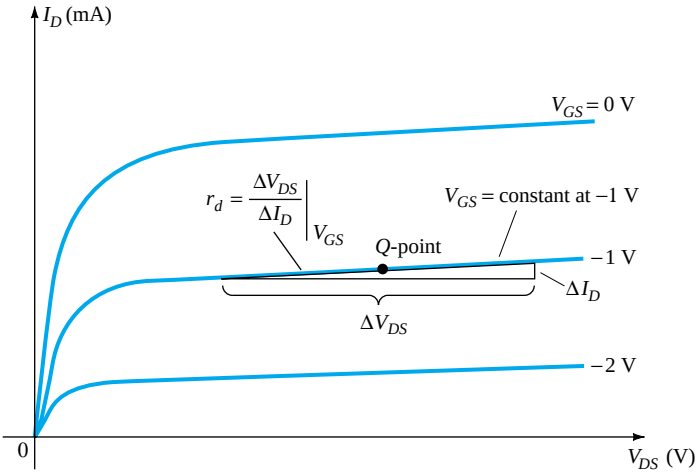


Figure 9.6 Definition of r_d using FET drain characteristics.

Note the requirement when applying Eq. (9.12) that the voltage V_{GS} remain constant when r_d is determined. This is accomplished by drawing a straight line approximating the V_{GS} line at the point of operation. A ΔV_{DS} or ΔI_D is then chosen and the other quantity measured off for use in the equation.

EXAMPLE 9.5

Determine the output impedance for the FET of Fig. 9.7 for $V_{GS} = 0$ V and $V_{GS} = -2$ V at $V_{DS} = 8$ V.

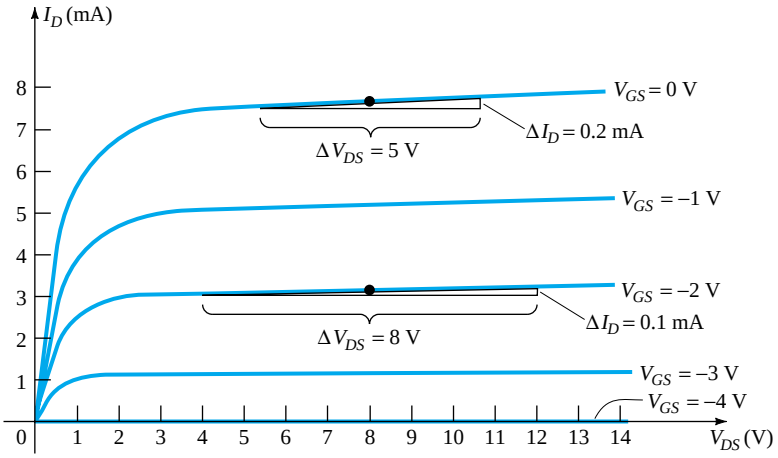


Figure 9.7 Drain characteristics used to calculate r_d in Example 9.5.

Solution

For $V_{GS} = 0$ V, a tangent line is drawn and ΔV_{DS} is chosen as 5 V, resulting in a ΔI_D of 0.2 mA. Substituting into Eq. (9.12),

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D}\bigg|_{V_{GS}=0\text{ V}} = \frac{5\text{ V}}{0.2\text{ mA}} = \mathbf{25\ k\Omega}$$

For $V_{GS} = -2$ V, a tangent line is drawn and ΔV_{DS} is chosen as 8 V, resulting in a ΔI_D of 0.1 mA. Substituting into Eq. (9.12),

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D}\bigg|_{V_{GS}=-2\text{ V}} = \frac{8\text{ V}}{0.1\text{ mA}} = \mathbf{80\ k\Omega}$$

revealing that r_d does change from one operating region to another, with lower values typically occurring at lower levels of V_{GS} (closer to 0 V).

FET AC Equivalent Circuit

Now that the important parameters of an ac equivalent circuit have been introduced and discussed, a model for the FET transistor in the ac domain can be constructed. The control of I_d by V_{gs} is included as a current source $g_m V_{gs}$ connected from drain to source as shown in Fig. 9.8. The current source has its arrow pointing from drain to source to establish a 180° phase shift between output and input voltages as will occur in actual operation.

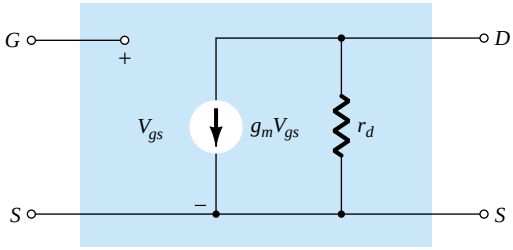


Figure 9.8 FET ac equivalent circuit.

The input impedance is represented by the open circuit at the input terminals and the output impedance by the resistor r_d from drain to source. Note that the gate to source voltage is now represented by V_{gs} (lower-case subscripts) to distinguish it from dc levels. In addition, take note of the fact that the source is common to both input and output circuits while the gate and drain terminals are only in "touch" through the controlled current source $g_m V_{gs}$.

In situations where r_d is ignored (assumed sufficiently large to other elements of the network to be approximated by an open circuit), the equivalent circuit is simply a current source whose magnitude is controlled by the signal V_{gs} and parameter g_m—clearly a voltage-controlled device.

EXAMPLE 9.6

Given $y_{fs} = 3.8$ mS and $y_{os} = 20\ \mu$S, sketch the FET ac equivalent model.

Solution

$$g_m = y_{fs} = 3.8\text{ mS} \quad \text{and} \quad r_d = \frac{1}{y_{os}} = \frac{1}{20\ \mu\text{S}} = 50\text{ k}\Omega$$

resulting in the ac equivalent model of Fig. 9.9.

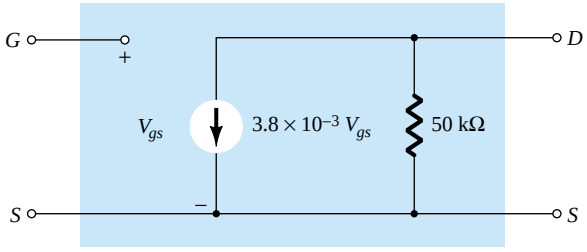


Figure 9.9 FET ac equivalent model for Example 9.6.

9.3 JFET FIXED-BIAS CONFIGURATION

Now that the FET equivalent circuit has been defined, a number of fundamental FET small-signal configurations will be investigated. The approach will parallel the ac analysis of BJT amplifiers with a determination of the important parameters of Z_i, Z_o, and A_v for each configuration.

The *fixed-bias* configuration of Fig. 9.10 includes the coupling capacitors C_1 and C_2 that isolate the dc biasing arrangement from the applied signal and load; they act as short-circuit equivalents for the ac analysis.

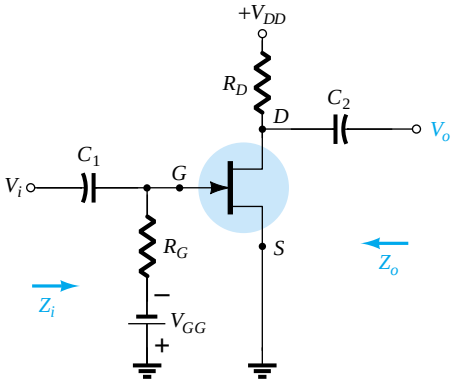


Figure 9.10 JFET fixed-bias configuration.

Once the level of g_m and r_d are determined from the dc biasing arrangement, specification sheet, or characteristics, the ac equivalent model can be substituted between the appropriate terminals as shown in Fig. 9.11. Note that both capacitors have the short-circuit equivalent because the reactance $X_C = 1/(2\pi fC)$ is sufficiently small compared to other impedance levels of the network, and the dc batteries V_{GG} and V_{DD} are set to zero volts by a short-circuit equivalent.

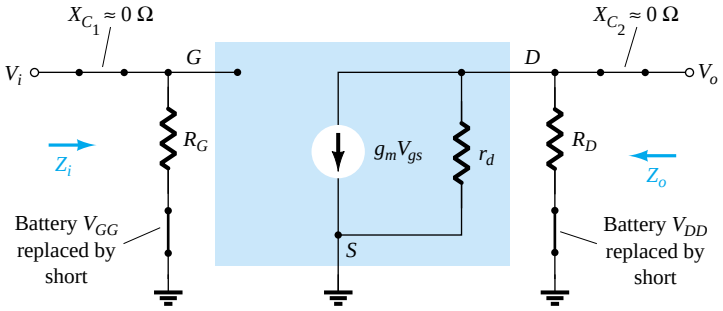


Figure 9.11 Substituting the JFET ac equivalent circuit unit into the network of Fig. 9.10.

The network of Fig. 9.11 is then carefully redrawn as shown in Fig. 9.12. Note the defined polarity of V_{gs}, which defines the direction of $g_m V_{gs}$. If V_{gs} is negative, the direction of the current source reverses. The applied signal is represented by V_i and the output signal across R_D by V_o.

***Z_i*:** Figure 9.12 clearly reveals that

$$Z_i = R_G \tag{9.13}$$

because of the open-circuit equivalence at the input terminals of the JFET.

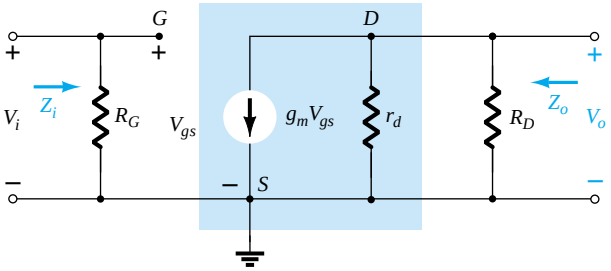


Figure 9.12 Redrawn network of Fig. 9.11.

Z_o: Setting $V_i = 0$ V as required by the definition of Z_o will establish V_{gs} as 0 V also. The result is $g_m V_{gs} = 0$ mA, and the current source can be replaced by an open-circuit equivalent as shown in Fig. 9.13. The output impedance is

$$Z_o = R_D \| r_d \tag{9.14}$$

If the resistance r_d is sufficiently large (at least 10:1) compared to R_D, the approximation $r_d \| R_D \cong R_D$ can often be applied and

$$Z_o \cong R_D \quad r_d \geq 10R_D \tag{9.15}$$

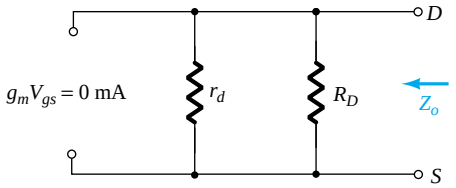


Figure 9.13 Determining Z_o.

A_v: Solving for V_o in Fig. 9.12, we find

$$V_o = -g_m V_{gs}(r_d \| R_D)$$

but $$V_{gs} = V_i$$

and $$V_o = -g_m V_i (r_d \| R_D)$$

so that

$$A_v = \frac{V_o}{V_i} = -g_m(r_d \| R_D) \tag{9.16}$$

If $r_d \geq 10R_D$:

$$A_v = \frac{V_o}{V_i} = -g_m R_D \quad r_d \geq 10R_D \tag{9.17}$$

Phase Relationship: The negative sign in the resulting equation for A_v clearly reveals a phase shift of 180° between input and output voltages.

EXAMPLE 9.7

The fixed-bias configuration of Example 6.1 had an operating point defined by $V_{GS_Q} = -2$ V and $I_{D_Q} = 5.625$ mA, with $I_{DSS} = 10$ mA and $V_P = -8$ V. The network is redrawn as Fig. 9.14 with an applied signal V_i. The value of y_{os} is provided as 40 μS.

(a) Determine g_m.
(b) Find r_d.
(c) Determine Z_i.
(d) Calculate Z_o.
(e) Determine the voltage gain A_v.
(f) Determine A_v ignoring the effects of r_d.

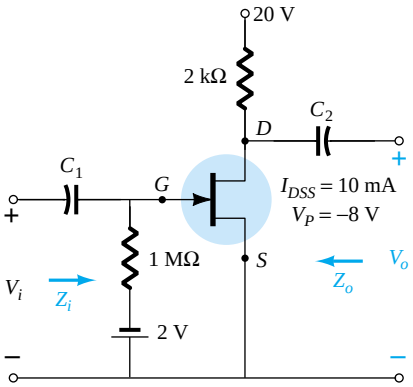


Figure 9.14 JFET configuration for Example 9.7.

Solution

(a) $$g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(10\text{ mA})}{8\text{ V}} = 2.5\text{ mS}$$

$$g_m = g_{m0}\left(1 - \frac{V_{GS_Q}}{V_P}\right) = 2.5\text{ mS}\left(1 - \frac{(-2\text{ V})}{(-8\text{ V})}\right) = \mathbf{1.88\ mS}$$

(b) $$r_d = \frac{1}{y_{os}} = \frac{1}{40\ \mu\text{S}} = \mathbf{25\ k\Omega}$$

(c) $Z_i = R_G = \mathbf{1\ M\Omega}$

(d) $Z_o = R_D \| r_d = 2\text{ k}\Omega \| 25\text{ k}\Omega = \mathbf{1.85\ k\Omega}$

(e) $A_v = -g_m(R_D \| r_d) = -(1.88\text{ mS})(1.85\text{ k}\Omega)$
$= \mathbf{-3.48}$

(f) $A_v = -g_m R_D = -(1.88\text{ mS})(2\text{ k}\Omega) = \mathbf{-3.76}$

As demonstrated in part (f), a ratio of 25 kΩ:2 kΩ = 12.5:1 between r_d and R_D resulted in a difference of 8% in solution.

9.4 JFET SELF-BIAS CONFIGURATION

Bypassed R_S

The fixed-bias configuration has the distinct disadvantage of requiring two dc voltage sources. The *self-bias* configuration of Fig. 9.15 requires only one dc supply to establish the desired operating point.

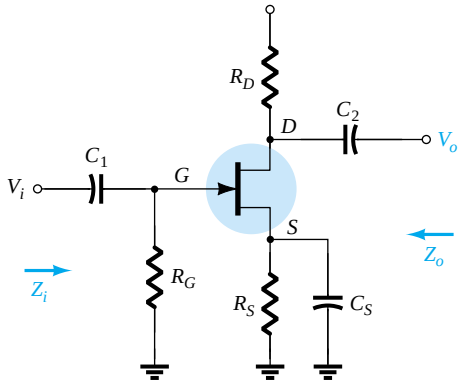


Figure 9.15 Self-bias JFET configuration.

The capacitor C_S across the source resistance assumes its short-circuit equivalence for dc, allowing R_S to define the operating point. Under ac conditions, the capacitor assumes the short-circuit state and "short circuits" the effects of R_S. If left in the ac, gain will be reduced as will be shown in the paragraphs to follow.

The JFET equivalent circuit is established in Fig. 9.16 and carefully redrawn in Fig. 9.17.

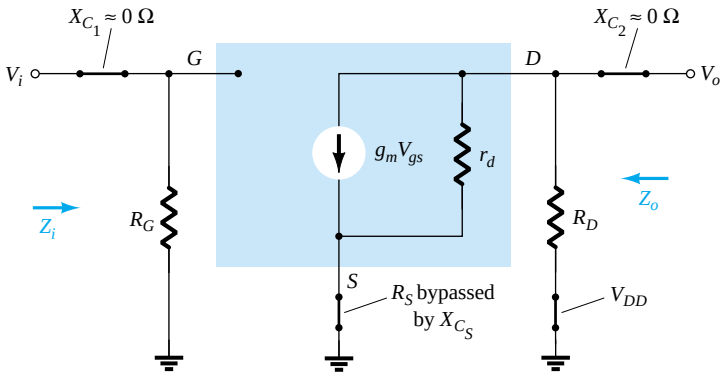


Figure 9.16 Network of Fig. 9.15 following the substitution of the JFET ac equivalent circuit.

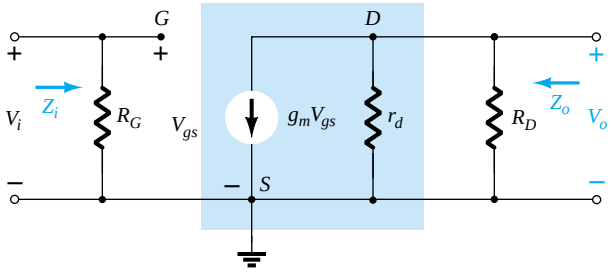


Figure 9.17 Redrawn network of Fig. 9.16.

Since the resulting configuration is the same as appearing in Fig. 9.12, the resulting equations Z_i, Z_o, and A_v will be the same.

Z_i:

$$Z_i = R_G \tag{9.18}$$

Z_o:

$$Z_o = r_d \| R_D \tag{9.19}$$

If $r_d \geq 10R_D$,

$$Z_o \cong R_D \Big|_{r_d \geq 10R_D} \tag{9.20}$$

A_v:

$$A_v = -g_m(r_d \| R_D) \tag{9.21}$$

If $r_d \geq 10R_D$,

$$A_v = -g_m R_D \Big|_{r_d \geq 10R_D} \tag{9.22}$$

Phase relationship: The negative sign in the solutions for A_v again indicates a phase shift of 180° between V_i and V_o.

Unbypassed R_S

If C_S is removed from Fig 9.15, the resistor R_S will be part of the ac equivalent circuit as shown in Fig. 9.18. In this case, there is no obvious way to reduce the network to lower its level of complexity. In determining the levels of Z_i, Z_o, and A_v, one must simply be very careful with notation and defined polarities and direction. Initially, the resistance r_d will be left out of the analysis to form a basis for comparison.

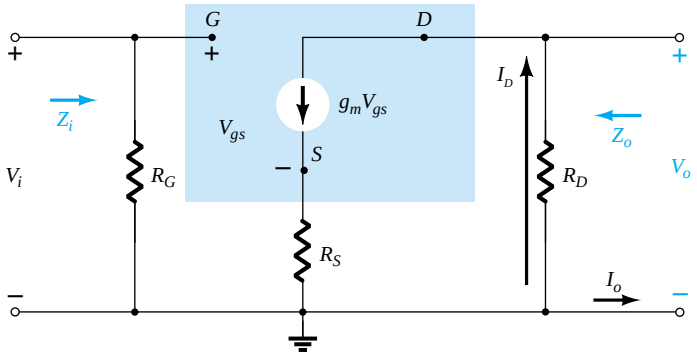


Figure 9.18 Self-bias JFET configuration including the effects of R_S with $r_d = \infty\Omega$.

Z_i: Due to the open-circuit condition between the gate and output network, the input remains the following:

$$Z_i = R_G \tag{9.23}$$

Z_o: The output impedance is defined by

$$Z_o = \frac{V_o}{I_o}\Big|_{V_i = 0}$$

Setting $V_i = 0$ V in Fig. 9.18 will result in the gate terminal being at ground potential (0 V). The voltage across R_G is then 0 V, and R_G has been effectively "shorted out" of the picture.

Applying Kirchhoff's current law will result in:

$$I_o + I_D = g_m V_{gS}$$

with $$V_{gs} = -(I_o + I_D)R_S$$

so that $$I_o + I_D = -g_m\,(I_o + I_D)R_S = -g_m I_o R_S - g_m I_D R_S$$

or $$I_o[1 + g_m R_S] = -I_D[1 + g_m R_S]$$

and $I_o = -I_D$ (the controlled current source $g_m V_{gs} = 0$ A for the applied conditions)

Since $$V_o = -I_D R_D$$

then $$V_o = -(-I_o)R_D = I_o R_D$$

and
$$\boxed{Z_o = \frac{V_o}{I_o} = R_D}\Big|_{r_d = \infty\Omega} \qquad (9.24)$$

If r_d is included in the network, the equivalent will appear as shown in Fig. 9.19.

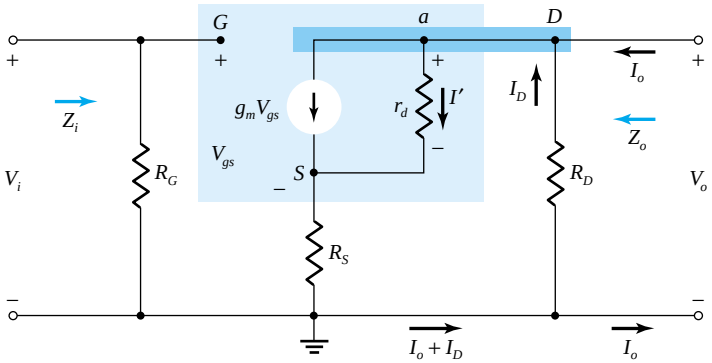


Figure 9.19 Including the effects of r_d in the self-bias JFET configuration.

Since $$Z_o = \frac{V_o}{I_o}\bigg|_{V_i = 0\text{ V}} = -\frac{I_D R_D}{I_o}$$

we should try to find an expression for I_o in terms of I_D.

Applying Kirchhoff's current law:

$$I_o = g_m V_{gs} + I_{r_d} - I_D$$

but $$V_{r_d} = V_o + V_{gs}$$

and $$I_o = g_m V_{gs} + \frac{V_o + V_{gs}}{r_d} - I_D$$

or $$I_o = \left(g_m + \frac{1}{r_d}\right) V_{gs} - \frac{I_D R_D}{r_d} - I_D \text{ using } V_o = -I_D R_D$$

Now, $$V_{gs} = -(I_D + I_o)\,R_S$$

so that $$I_o = -\left(g_m + \frac{1}{r_d}\right)(I_D + I_o)\,R_S - \frac{I_D R_D}{r_d} - I_D$$

with the result that $$I_o\left[1 + g_m R_s + \frac{R_S}{r_d}\right] = -I_D\left[1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d}\right]$$

or $$I_o = \frac{-I_D\left[1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d}\right]}{1 + g_m R_S + \frac{R_S}{r_d}}$$

and
$$Z_o = \frac{V_o}{I_o} = \frac{-I_D R_D}{-I_D\left(1 + g_m R_s + \frac{R_s}{r_d} + \frac{R_D}{r_d}\right)}$$
$$1 + g_m R_S + \frac{R_S}{r_d}$$

and finally,
$$\boxed{Z_o = \frac{\left[1 + g_m R_S + \frac{R_S}{r_d}\right]}{\left[1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d}\right]} R_D} \tag{9.25a}$$

For $r_d \geq 10\ R_D$, $\left(1 + g_m R_S + \frac{R_S}{r_d}\right) >> \frac{R_D}{r_d}$ and $1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d}$ $\cong 1 + g_m R_S + \frac{R_S}{r_d}$ and

$$\boxed{Z_o = R_D}_{r_d \geq 10R_D} \tag{9.25b}$$

A_v: For the network of Fig. 9.19, an application of Kirchhoff's voltage law on the input circuit will result in

$$V_i - V_{gs} - V_{R_S} = 0$$
$$V_{gs} = V_i - I_D R_S$$

The voltage across r_d using Kirchhoff's voltage law is

$$V_o - V_{R_S}$$

and
$$I' = \frac{V_o - V_{R_S}}{r_d}$$

so that an application of Kirchhoff's current law will result in

$$I_D = g_m V_{gs} + \frac{V_o - V_{R_S}}{r_d}$$

Substituting for V_{gs} from above and substituting for V_o and V_{R_S} we have

$$I_D = g_m[V_i - I_D R_S] + \frac{(-I_D R_D) - (I_D R_S)}{r_d}$$

so that
$$I_D\left[1 + g_m R_S + \frac{R_D + R_S}{r_d}\right] = g_m V_i$$

or
$$I_D = \frac{g_m V_i}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}$$

The output voltage is then

$$V_o = -I_D R_D = -\frac{g_m R_D V_i}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}$$

and
$$\boxed{A_v = \frac{V_o}{V_i} = -\frac{g_m R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}} \tag{9.26}$$

Again, if $r_d \geq 10(R_D + R_S)$,

$$A_v = \frac{V_o}{V_i} = -\frac{g_m R_D}{1 + g_m R_S} \qquad (9.27)$$

$r_d \geq 10(R_D + R_S)$

Phase Relationship: The negative sign in Eq. (9.26) again reveals that a 180° phase shift will exist between V_i and V_o.

EXAMPLE 9.8

The self-bias configuration of Example 6.2 has an operating point defined by $V_{GS_Q} = -2.6$ V and $I_{D_Q} = 2.6$ mA, with $I_{DSS} = 8$ mA and $V_P = -6$ V. The network is redrawn as Fig. 9.20 with an applied signal V_i. The value of y_{os} is given as 20 μS.

(a) Determine g_m.
(b) Find r_d.
(c) Find Z_i.
(d) Calculate Z_o with and without the effects of r_d. Compare the results.
(e) Calculate A_v with and without the effects of r_d. Compare the results.

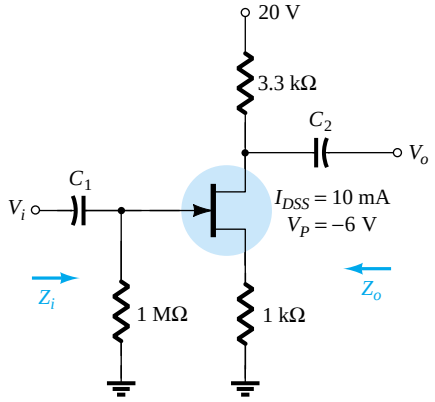


Figure 9.20 Network for Example 9.8.

Solution

(a) $g_{m0} = \dfrac{2I_{DSS}}{|V_P|} = \dfrac{2(8\text{ mA})}{6\text{ V}} = 2.67\text{ mS}$

$$g_m = g_{m0}\left(1 - \frac{V_{GS_Q}}{V_P}\right) = 2.67\text{ mS}\left(1 - \frac{(-2.6\text{ V})}{(-6\text{ V})}\right) = \mathbf{1.51\ mS}$$

(b) $r_d = \dfrac{1}{y_{os}} = \dfrac{1}{20\ \mu\text{S}} = \mathbf{50\ k\Omega}$

(c) $Z_i = R_G = \mathbf{1\ M\Omega}$

(d) With r_d:

$$r_d = 50\text{ k}\Omega > 10\,R_D = 33\text{ k}\Omega$$

Therefore,

$$Z_o = R_D = \mathbf{3.3\ k\Omega}$$

If $r_d = \infty\ \Omega$

$$Z_o = R_D = \mathbf{3.3\ k\Omega}$$

(e) With r_d:

$$A_v = \frac{-g_m R_D}{1 + g_m R_S + \dfrac{R_D + R_S}{r_d}} = \frac{-(1.51\text{ mS})(3.3\text{ k}\Omega)}{1 + (1.51\text{ mS})(1\text{ k}\Omega) + \dfrac{3.3\text{ k}\Omega + 1\text{ k}\Omega}{50\text{ k}\Omega}}$$

$$= \mathbf{-1.92}$$

Without r_d:

$$A_v = \frac{-g_m R_D}{1 + g_m R_S} = \frac{-(1.51\ \text{mS})(3.3\ \text{k}\Omega)}{1 + (1.51\ \text{mS})(1\ \text{k}\Omega)} = \mathbf{-1.98}$$

As above, the effect of r_d was minimal because the condition $r_d \geq 10(R_D + R_S)$ was satisfied.

Note also that the typical gain of a JFET amplifier is less than that generally encountered for BJTs of similar configurations. Keep in mind, however, that Z_i is magnitudes greater than the typical Z_i of a BJT, which will have a very positive effect on the overall gain of a system.

9.5 JFET VOLTAGE-DIVIDER CONFIGURATION

The popular voltage-divider configuration for BJTs can also be applied to JFETs as demonstrated in Fig. 9.21.

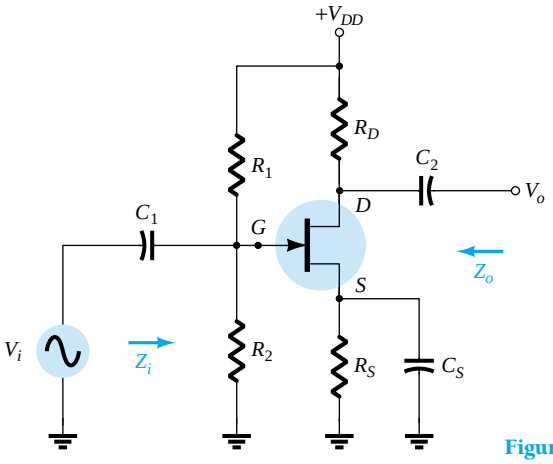


Figure 9.21 JFET voltage-divider configuration.

Substituting the ac equivalent model for the JFET will result in the configuration of Fig. 9.22. Replacing the dc supply V_{DD} by a short-circuit equivalent has grounded one end of R_1 and R_D. Since each network has a common ground, R_1 can be brought down in parallel with R_2 as shown in Fig. 9.23. R_D can also be brought down to ground but in the output circuit across r_d. The resulting ac equivalent network now has the basic format of some of the networks already analyzed.

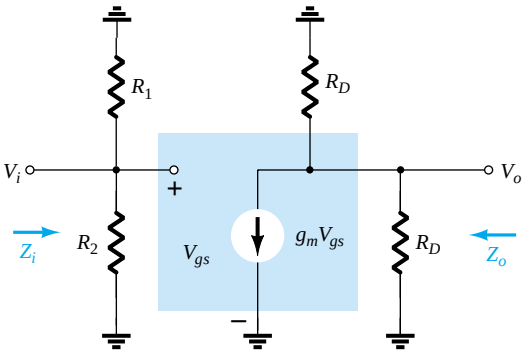


Figure 9.22 Network of Fig. 9.21 under ac conditions.

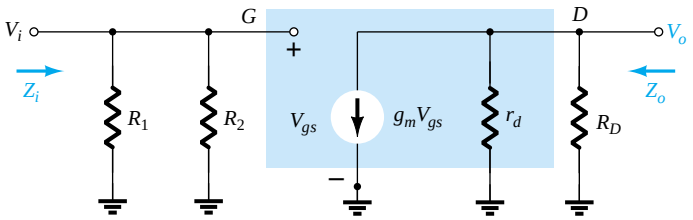


Figure 9.23 Redrawn network of Fig. 9.22.

Z_i: R_1 and R_2 are in parallel with the open-circuit equivalence of the JFET resulting in

$$Z_i = R_1 \| R_2 \tag{9.28}$$

Z_o: Setting $V_i = 0$ V will set V_{gs} and $g_m V_{gs}$ to zero and

$$Z_o = r_d \| R_D \tag{9.29}$$

For $r_d \geq 10R_D$,

$$Z_o \cong R_D \Big|_{r_d \geq 10R_D} \tag{9.30}$$

A_v:

$$V_{gs} = V_i$$

and

$$V_o = -g_m V_{gs}(r_d \| R_D)$$

so that

$$A_v = \frac{V_o}{V_i} = \frac{-g_m V_{gs}(r_d \| R_D)}{V_{gs}}$$

and

$$A_v = \frac{V_o}{V_i} = -g_m(r_d \| R_D) \tag{9.31}$$

If $r_d \geq 10R_D$,

$$A_v = \frac{V_o}{V_i} \cong -g_m R_D \Big|_{r_d \geq 10R_D} \tag{9.32}$$

Note that the equations for Z_o and A_v are the same as obtained for the fixed-bias and self-bias (with bypassed R_S) configurations. The only difference is the equation for Z_i, which is now sensitive to the parallel combination of R_1 and R_2.

9.6 JFET SOURCE-FOLLOWER (COMMON-DRAIN) CONFIGURATION

The JFET equivalent of the BJT emitter-follower configuration is the source-follower configuration of Fig. 9.24. Note that the output is taken off the source terminal and, when the dc supply is replaced by its short-circuit equivalent, the drain is grounded (hence, the terminology common-drain).

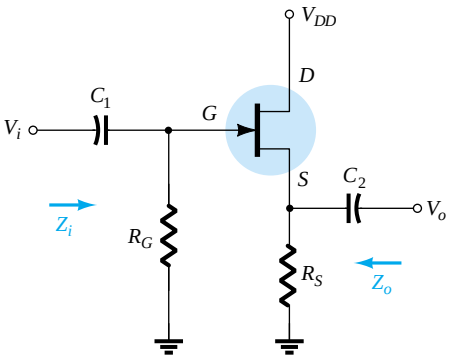


Figure 9.24 JFET source-follower configuration.

Substituting the JFET equivalent circuit will result in the configuration of Fig. 9.25. The controlled source and internal output impedance of the JFET are tied to ground at one end and R_S on the other, with V_o across R_S. Since g_mV_{gs}, r_d, and R_S are connected to the same terminal and ground, they can all be placed in parallel as shown in Fig. 9.26. The current source reversed direction but V_{gs} is still defined between the gate and source terminals.

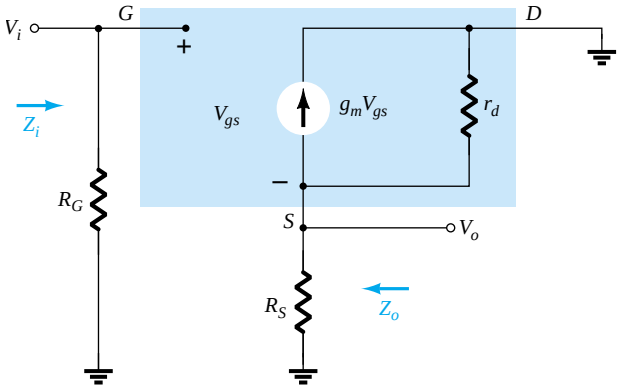


Figure 9.25 Network of Fig. 9.24 following the substitution of the JFET ac equivalent model.

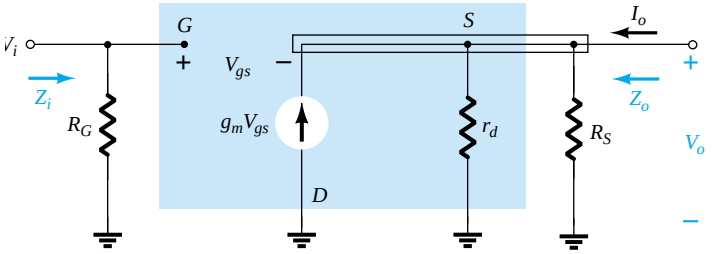


Figure 9.26 Network of Fig. 9.25 redrawn.

Z_i: Figure 9.26 clearly reveals that Z_i is defined by

$$Z_i = R_G \tag{9.33}$$

Z_o: Setting $V_i = 0$ V will result in the gate terminal being connected directly to ground as shown in Fig. 9.27. The fact that V_{gs} and V_o are across the same parallel network results in $V_o = -V_{gs}$.

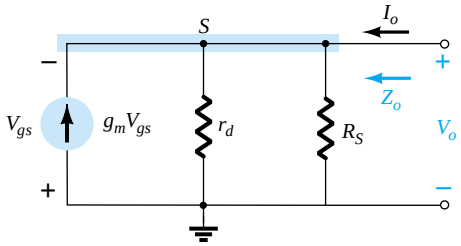


Figure 9.27 Determining Z_o for the network of Fig. 9.24.

Applying Kirchhoff's current law at node *s*,

$$I_o + g_mV_{gs} = I_{r_d} + I_{R_S}$$
$$= \frac{V_o}{r_d} + \frac{V_o}{R_S}$$

The result is

$$I_o = V_o\left[\frac{1}{r_d} + \frac{1}{R_S}\right] - g_mV_{gs}$$
$$= V_o\left[\frac{1}{r_d} + \frac{1}{R_S}\right] - g_m[-V_o]$$
$$= V_o\left[\frac{1}{r_d} + \frac{1}{R_S} + g_m\right]$$

and $$Z_o = \frac{V_o}{I_o} = \frac{V_o}{V_o\left[\frac{1}{r_d} + \frac{1}{R_S} + g_m\right]} = \frac{1}{\frac{1}{r_d} + \frac{1}{R_S} + g_m} = \frac{1}{\frac{1}{r_d} + \frac{1}{R_S} + \frac{1}{1/g_m}}$$

which has the same format as the total resistance of three parallel resistors. Therefore,

$$\boxed{Z_o = r_d \| R_S \| 1/g_m} \tag{9.34}$$

For $r_d \geq 10R_S$,

$$\boxed{Z_o \cong R_S \| 1/g_m}_{r_d \geq 10R_S} \tag{9.35}$$

A_v: The output voltage V_o is determined by

$$V_o = g_m V_{gs}(r_d \| R_S)$$

and applying Kirchhoff's voltage law around the perimeter of the network of Fig. 9.26 will result in

$$V_i = V_{gs} + V_o$$

and $$V_{gs} = V_i - V_o$$

so that $$V_o = g_m(V_i - V_o)(r_d \| R_S)$$

or $$V_o = g_m V_i(r_d \| R_S) - g_m V_o(r_d \| R_S)$$

and $$V_o[1 + g_m(r_d \| R_S)] = g_m V_i(r_d \| R_S)$$

so that $$\boxed{A_v = \frac{V_o}{V_i} = \frac{g_m(r_d \| R_S)}{1 + g_m(r_d \| R_S)}} \tag{9.36}$$

In the absence of r_d or if $r_d \geq 10R_S$,

$$\boxed{A_v = \frac{V_o}{V_i} \cong \frac{g_m R_S}{1 + g_m R_S}}_{r_d \geq 10R_S} \tag{9.37}$$

Since the bottom of Eq. (9.36) is larger than the numerator by a factor of one, the gain can never be equal to or greater than one (as encountered for the emitter-follower BJT network).

Phase Relationship: Since A_v of Eq. (9.36) is a positive quantity, V_o and V_i are in phase for the JFET source-follower configuration.

EXAMPLE 9.9

A dc analysis of the source-follower network of Fig. 9.28 will result in $V_{GS_Q} = -2.86$ V and $I_{D_Q} = 4.56$ mA.

(a) Determining g_m.
(b) Find r_d.
(c) Determine Z_i.
(d) Calculate Z_o with and without r_d. Compare results.
(e) Determine A_v with and without r_d. Compare results.

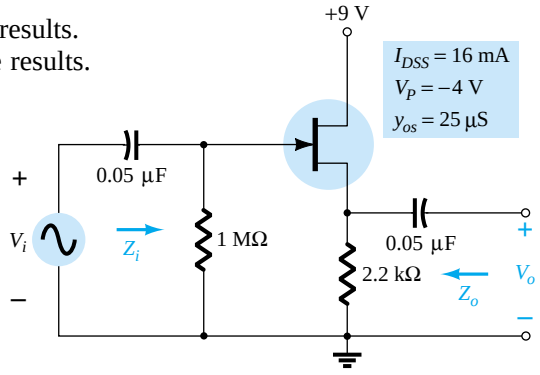


Figure 9.28 Network to be analyzed in Example 9.9.

Solution

(a) $g_{m0} = \dfrac{2I_{DSS}}{|V_P|} = \dfrac{2(16\text{ mA})}{4\text{ V}} = 8\text{ mS}$

$$g_m = g_{m0}\left(1 - \frac{V_{GS_Q}}{V_P}\right) = 8\text{ mS}\left(1 - \frac{(-2.86\text{ V})}{(-4\text{ V})}\right) = \mathbf{2.28\ mS}$$

(b) $r_d = \dfrac{1}{y_{os}} = \dfrac{1}{25\ \mu\text{S}} = \mathbf{40\ k\Omega}$

(c) $Z_i = R_G = \mathbf{1\ M\Omega}$

(d) With r_d:

$$\begin{aligned} Z_o = r_d \| R_S \| 1/g_m &= 40\text{ k}\Omega \| 2.2\text{ k}\Omega \| 1/2.28\text{ mS} \\ &= 40\text{ k}\Omega \| 2.2\text{ k}\Omega \| 438.6\ \Omega \\ &= \mathbf{362.52\ \Omega} \end{aligned}$$

revealing that Z_o is often relatively small and determined primarily by $1/g_m$. Without r_d:

$$Z_o = R_S \| 1/g_m = 2.2\text{ k}\Omega \| 438.6\ \Omega = \mathbf{365.69\ \Omega}$$

revealing that r_d typically has little impact on Z_o.

(e) With r_d:

$$A_v = \frac{g_m(r_d \| R_S)}{1 + g_m(r_d \| R_S)} = \frac{(2.28\text{ mS})(40\text{ k}\Omega \| 2.2\text{ k}\Omega)}{1 + (2.28\text{ mS})(40\text{ k}\Omega \| 2.2\text{ k}\Omega)}$$

$$\frac{(2.28\text{ mS})(2.09\text{ k}\Omega)}{1 + (2.28\text{ mS})(2.09\text{ k}\Omega)} = \frac{4.77}{1 + 4.77} = \mathbf{0.83}$$

which is less than 1 as predicted above.
Without r_d:

$$A_v = \frac{g_m R_S}{1 + g_m R_S} = \frac{(2.28\text{ mS})(2.2\text{ k}\Omega)}{1 + (2.28\text{ mS})(2.2\text{ k}\Omega)}$$

$$= \frac{5.02}{1 + 5.02} = \mathbf{0.83}$$

revealing that r_d usually has little impact on the gain of the configuration.

9.7 JFET COMMON-GATE CONFIGURATION

The last JFET configuration to be analyzed in detail is the common-gate configuration of Fig. 9.29, which parallels the common-base configuration employed with BJT transistors.

Substituting the JFET equivalent circuit will result in Fig. 9.30. Note the continuing requirement that the controlled source $g_m V_{gs}$ be connected from drain to source with r_d in parallel. The isolation between input and output circuits has obviously been lost since the gate terminal is now connected to the common ground of the network. In addition, the resistor connected between input terminals is no longer R_G but the resistor R_S connected from source to ground. Note also the location of the controlling voltage V_{gs} and the fact that it appears directly across the resistor R_S.

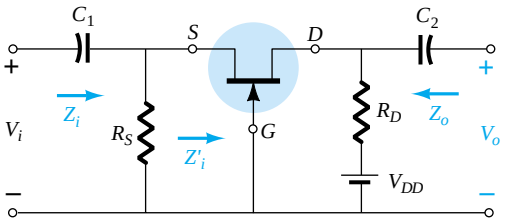


Figure 9.29 JFET common-gate configuration.

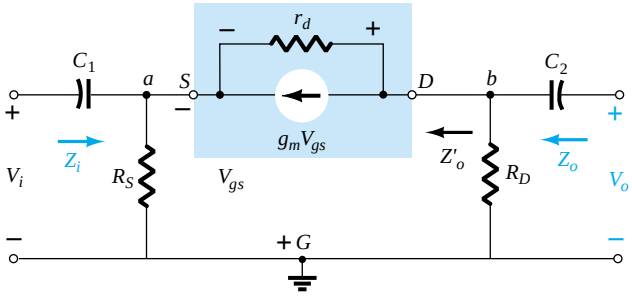


Figure 9.30 Network of Fig. 9.29 following substitution of JFET ac equivalent model.

***Z*$_i$:** The resistor R_S is directly across the terminals defining Z_i. Let us therefore find the impedance Z'_i of Fig. 9.29, which will simply be in parallel with R_S when Z_i is defined.

The network of interest is redrawn as Fig. 9.31. The voltage $V' = -V_{gs}$. Applying Kirchhoff's voltage law around the output perimeter of the network will result in

$$V' - V_{r_d} - V_{R_D} = 0$$

and

$$V_{r_d} = V' - V_{R_D} = V' - I'R_D$$

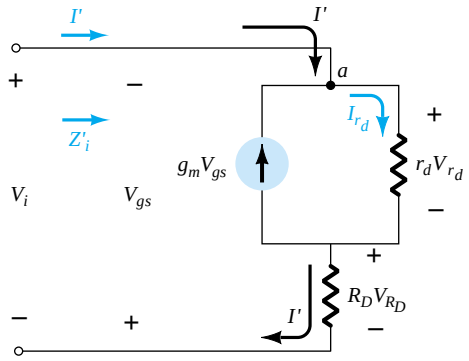


Figure 9.31 Determining Z'_i for the network of Fig. 9.29.

Applying Kirchhoff's current law at node a results in

$$I' + g_m V_{gs} = I_{r_d}$$

and

$$I' = I_{r_d} - g_m V_{gs} = \frac{(V' - I'R_D)}{r_d} - g_m V_{gs}$$

or

$$I' = \frac{V'}{r_d} - \frac{I'R_D}{r_d} - g_m[-V']$$

so that

$$I'\left[1 + \frac{R_D}{r_d}\right] = V'\left[\frac{1}{r_d} + g_m\right]$$

and

$$Z'_i = \frac{V'}{I'} = \frac{\left[1 + \dfrac{R_D}{r_d}\right]}{\left[g_m + \dfrac{1}{r_d}\right]} \tag{9.38}$$

or

$$Z'_i = \frac{V'}{I'} = \frac{r_d + R_D}{1 + g_m r_d}$$

and $$Z_i = R_S \| Z_i'$$

results in $$\boxed{Z_i = R_S \| \left[\frac{r_d + R_D}{1 + g_m r_d} \right]} \tag{9.39}$$

If $r_d \geq 10R_D$, Eq. (9.38) permits the following approximation since $R_D/r_d \ll 1$ and $1/r_d \ll g_m$:

$$Z_i' = \frac{\left[1 + \frac{R_D}{r_d}\right]}{\left[g_m + \frac{1}{r_d}\right]} \cong \frac{1}{g_m}$$

and $$\boxed{Z_i \cong R_S \| 1/g_m}_{r_d \geq 10R_D} \tag{9.40}$$

Z_o: Substituting $V_i = 0$ V in Fig. 9.30 will "short-out" the effects of R_S and set V_{gs} to 0 V. The result is $g_m V_{gs} = 0$, and r_d will be in parallel with R_D. Therefore,

$$\boxed{Z_o = R_D \| r_d} \tag{9.41}$$

For $r_d \geq 10R_D$,

$$\boxed{Z_o \cong R_D}_{r_d \geq 10R_D} \tag{9.42}$$

A_v: Figure 9.30 reveals that

$$V_i = -V_{gs}$$

and $$V_o = I_D R_D$$

The voltage across r_d is

$$V_{r_d} = V_o - V_i$$

and $$I_{r_d} = \frac{V_o - V_i}{r_d}$$

Applying Kirchhoff's current law at node b in Fig. 9.30 results in

$$I_{r_d} + I_D + g_m V_{gs} = 0$$

and $$I_D = -I_{rd} - g_m V_{gs}$$

$$= -\left[\frac{V_o - V_i}{r_d}\right] - g_m[-V_i]$$

$$I_D = \frac{V_i - V_o}{r_d} + g_m V_i$$

so that $$V_o = I_D R_D = \left[\frac{V_i - V_o}{r_d} + g_m V_i\right] R_D$$

$$= \frac{V_i R_D}{r_d} - \frac{V_o R_D}{r_d} + g_m$$

and $$V_o\left[1 + \frac{R_D}{r_d}\right] = V_i\left[\frac{R_D}{r_d} + g_m R_D\right]$$

with
$$A_v = \frac{V_o}{V_i} = \frac{\left[g_m R_D + \frac{R_D}{r_d}\right]}{\left[1 + \frac{R_D}{r_d}\right]} \tag{9.43}$$

For $r_d \geq 10R_D$, the factor R_D/r_d of Eq. (9.43) can be dropped as a good approximation and

$$A_v = g_m R_D \quad \text{for } r_d \geq 10R_D \tag{9.44}$$

Phase Relationship: The fact that A_v is a positive number will result in an *in-phase* relationship between V_o and V_i for the common-gate configuration.

EXAMPLE 9.10

Although the network of Fig. 9.32 may not initially appear to be of the common-gate variety, a close examination will reveal that it has all the characteristics of Fig. 9.29. If $V_{GS_Q} = -2.2$ V and $I_{D_Q} = 2.03$ mA:

(a) Determine g_m.
(b) Find r_d.
(c) Calculate Z_i with and without r_d. Compare results.
(d) Find Z_o with and without r_d. Compare results.
(e) Determine V_o with and without r_d. Compare results.

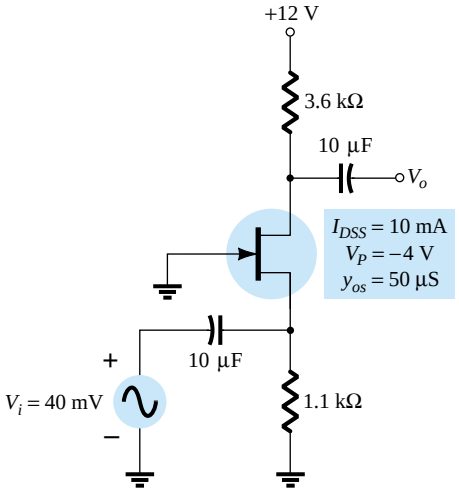


Figure 9.32 Network for Example 9.10.

Solution

(a) $$g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(10\text{ mA})}{4\text{ V}} = 5\text{ mS}$$

$$g_m = g_{m0}\left(1 - \frac{V_{GS_Q}}{V_P}\right) = 5\text{ mS}\left(1 - \frac{(-2.2\text{ V})}{(-4\text{ V})}\right) = \mathbf{2.25\ mS}$$

(b) $$r_d = \frac{1}{y_{os}} = \frac{1}{50\ \mu\text{S}} = \mathbf{20\ k\Omega}$$

(c) With r_d:

$$Z_i = R_S \| \left[\frac{r_d + R_D}{1 + g_m r_d}\right] = 1.1\text{ k}\Omega \| \left[\frac{20\text{ k}\Omega + 3.6\text{ k}\Omega}{1 + (2.25\text{ ms})(20\text{ k}\Omega)}\right]$$
$$= 1.1\text{ k}\Omega \| 0.51\text{ k}\Omega = \mathbf{0.35\ k\Omega}$$

Without r_d:

$$Z_i = R_S\|1/g_m = 1.1\ \text{k}\Omega\|1/2.25\ \text{ms} = 1.1\ \text{k}\Omega\|0.44\ \text{k}\Omega$$
$$= \mathbf{0.31\ k\Omega}$$

Even though the condition,

$$r_d \geq 10R_D = > 20\ \text{k}\Omega \geq 10(3.6\ \text{k}\Omega) = > 20\ \text{k}\Omega \geq 36\ \text{k}\Omega$$

is *not* satisfied, both equations result in essentially the same level of impedance. In this case, $1/g_m$ was the predominant factor.

(d) With r_d:

$$Z_o = R_D\|r_d = 3.6\ \text{k}\Omega\|20\ \text{k}\Omega = \mathbf{3.05\ k\Omega}$$

Without r_d:

$$Z_o = R_D = \mathbf{3.6\ k\Omega}$$

Again the condition $r_d \geq 10R_D$ is *not* satisfied, but both results are reasonably close. R_D is certainly the predominant factor in this example.

(e) With r_d:

$$A_v = \frac{\left[g_m R_D + \dfrac{R_D}{r_d}\right]}{\left[1 + \dfrac{R_D}{r_d}\right]} = \frac{\left[(2.25\ \text{mS})(3.6\ \text{k}\Omega) + \dfrac{3.6\ \text{k}\Omega}{20\ \text{k}\Omega}\right]}{\left[1 + \dfrac{3.6\ \text{k}\Omega}{20\ \text{k}\Omega}\right]}$$
$$= \frac{8.1 + 0.18}{1 + 0.18} = \mathbf{7.02}$$

and $$A_v = \frac{V_o}{V_i} = \blacktriangleright V_o = A_v V_i = (7.02)(40\ \text{mV}) = \mathbf{280.8\ mV}$$

Without r_d:

$$A_v = g_m R_D = (2.25\ \text{mS})(3.6\ \text{k}\Omega) = \mathbf{8.1}$$

with $$V_o = A_v V_i = (8.1)(40\ \text{mV}) = \mathbf{324\ mV}$$

In this case, the difference is a little more noticeable but not dramatically so.

Example 9.10 demonstrates that even though the condition $r_d \geq 10R_D$ was not satisfied, the results for the parameters given were not significantly different using the exact and approximate equations. In fact, in most cases, the approximate equations can be used to find a reasonable idea of particular levels with a reduced amount of effort.

9.8 DEPLETION-TYPE MOSFETs

The fact that Shockley's equation is also applicable to depletion-type MOSFETs results in the same equation for g_m. In fact, the ac equivalent model for D-MOSFETs is exactly the same as that employed for JFETs as shown in Fig. 9.33.

The only difference offered by D-MOSFETs is that V_{GS_Q} can be positive for *n*-channel devices and negative for *p*-channel units. The result is that g_m can be greater than g_{m0} as demonstrated by the example to follow. The range of r_d is very similar to that encountered for JFETs.

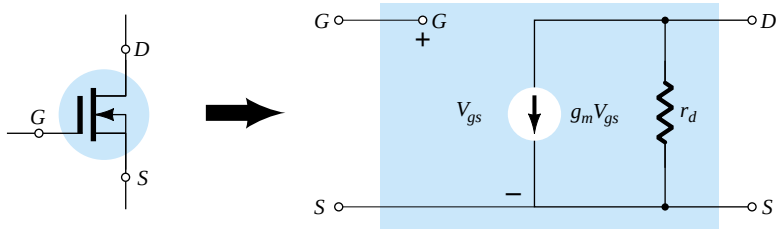


Figure 9.33 D-MOSFET ac equivalent model.

EXAMPLE 9.11

The network of Fig. 9.34 was analyzed as Example 6.8, resulting in $V_{GS_Q} = 0.35$ V and $I_{D_Q} = 7.6$ mA.

(a) Determine g_m and compare to g_{m0}.
(b) Find r_d.
(c) Sketch the ac equivalent network for Fig. 9.34.
(d) Find Z_i.
(e) Calculate Z_o.
(f) Find A_v.

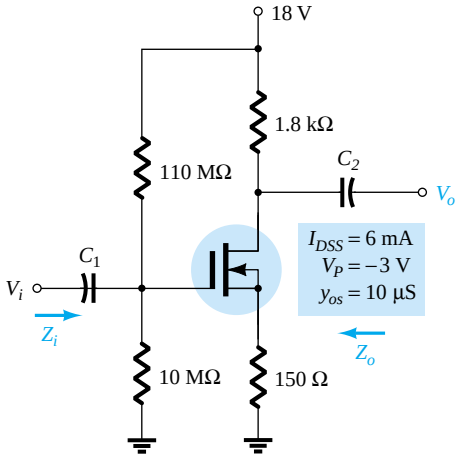


Figure 9.34 Network for Example 9.11.

Solution

(a) $$g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(6\text{ mA})}{3\text{ V}} = 4\text{ mS}$$

$$g_m = g_{m0}\left(1 - \frac{V_{GS_Q}}{V_P}\right) = 4\text{ mS}\left(1 - \frac{(+0.35\text{ V})}{(-3\text{ V})}\right) = 4\text{ mS}(1 + 0.117) = \mathbf{4.47\ mS}$$

(b) $$r_d = \frac{1}{y_{os}} = \frac{1}{10\ \mu S} = \mathbf{100\ k\Omega}$$

(c) See Fig. 9.35. Note the similarities with the network of Fig. 9.23. Equations (9.28) through (9.32) are therefore applicable.

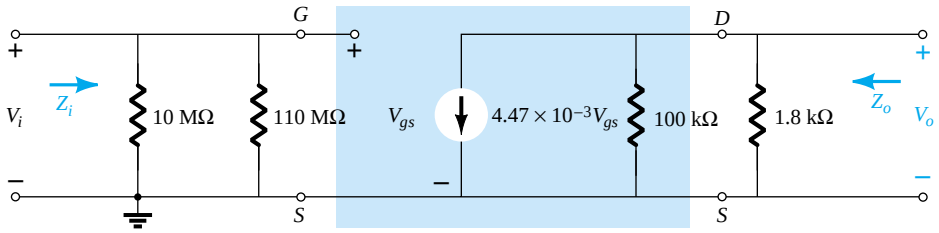


Figure 9.35 AC equivalent circuit for Fig. 9.34.

(d) Eq. (9.28): $Z_i = R_1 \| R_2 = 10\ \text{M}\Omega \| 110\ \text{M}\Omega = \mathbf{9.17\ M\Omega}$
(e) Eq. (9.29): $Z_o = r_d \| R_D = 100\ \text{k}\Omega \| 1.8\ \text{k}\Omega = \mathbf{1.77\ k\Omega} \cong R_D = \mathbf{1.8\ k\Omega}$
(f) $r_d \geq 10R_D \rightarrow 100\ \text{k}\Omega \geq 18\ \text{k}\Omega$

Eq. (9.32): $A_v = -g_m R_D = -(4.47\ \text{mS})(1.8\ \text{k}\Omega) = \mathbf{8.05}$

9.9 ENHANCEMENT-TYPE MOSFETs

The enhancement-type MOSFET can be either an *n*-channel (*n*MOS) or *p*-channel (*p*MOS) device, as shown in Fig. 9.36. The ac small-signal equivalent circuit of either device is shown in Fig. 9.36, revealing an open-circuit between gate and drain-source channel and a current source from drain to source having a magnitude dependent on the gate-to-source voltage. There is an output impedance from drain to source r_d, which is usually provided on specification sheets as an admittance y_{os}. The device transconductance, g_m, is provided on specification sheets as the forward transfer admittance, y_{fs}.

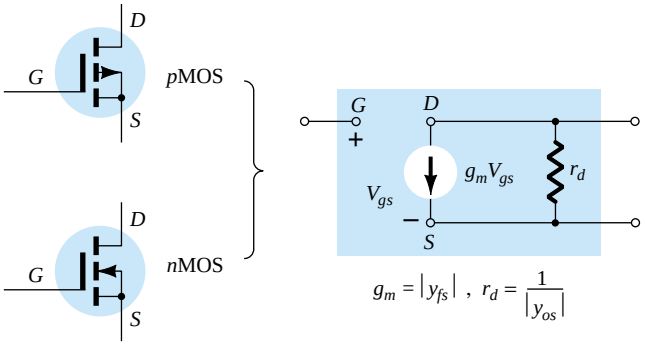


Figure 9.36 Enhancement MOSFET ac small-signal model.

In our analysis of JFETs, an equation for g_m was derived from Shockley's equation. For E-MOSFETs, the relationship between output current and controlling voltage is defined by

$$I_D = k(V_{GS} - V_{GS(\text{Th})})^2$$

Since g_m is still defined by

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

we can take the derivative of the transfer equation to determine g_m as an operating point. That is,

$$g_m = \frac{dI_D}{dV_{GS}} = \frac{d}{dV_{GS}} k(V_{GS} - V_{GS(\text{Th})})^2 = k\frac{d}{dV_{GS}}(V_{GS} - V_{GS(\text{Th})})^2$$

$$= 2k(V_{GS} - V_{GS(\text{Th})})\frac{d}{dV_{GS}}(V_{GS} - V_{GS(\text{Th})}) = 2k(V_{GS} - V_{GS(\text{Th})})(1 - 0)$$

and

$$\boxed{g_m = 2k(V_{GS_Q} - V_{GS(\text{Th})})} \qquad (9.45)$$

Recall that the constant k can be determined from a given typical operating point on a specification sheet. In every other respect, the ac analysis is the same as that employed for JFETs or D-MOSFETs. Be aware, however, that the characteristics of an E-MOSFET are such that the biasing arrangements are somewhat limited.

9.10 E-MOSFET DRAIN-FEEDBACK CONFIGURATION

The E-MOSFET drain-feedback configuration appears in Fig. 9.37. Recall from dc calculations that R_G could be replaced by a short-circuit equivalent since $I_G = 0$ A and therefore $V_{R_G} = 0$ V. However, for ac situations it provides an important high impedance between V_o and V_i. Otherwise, the input and output terminals would be connected directly and $V_o = V_i$.

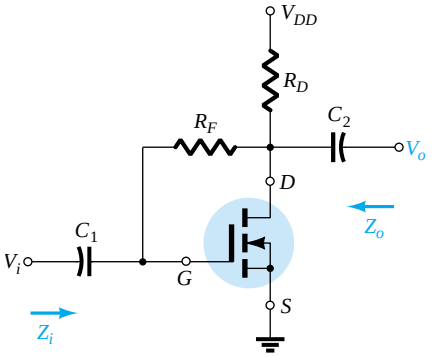


Figure 9.37 E-MOSFET drain-feedback configuration.

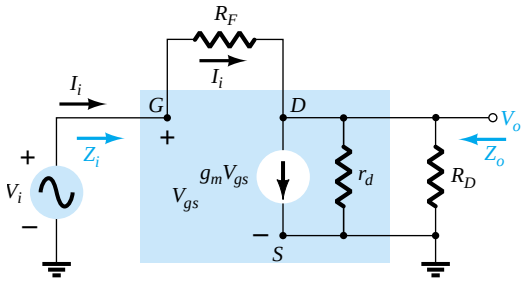


Figure 9.38 AC equivalent of the network of Fig. 9.37.

Substituting the ac equivalent model for the device will result in the network of Fig. 9.38. Note that R_F is not within the shaded area defining the equivalent model of the device but does provide a direct connection between input and output circuits.

Z_i: Applying Kirchhoff's current law to the output circuit (at node D in Fig. 9.38) results in

$$I_i = g_m V_{gs} + \frac{V_o}{r_d \| R_D}$$

and

$$V_{gs} = V_i$$

so that

$$I_i = g_m V_i + \frac{V_o}{r_d \| R_D}$$

or

$$I_i - g_m V_i = \frac{V_o}{r_d \| R_D}$$

Therefore,

$$V_o = (r_d \| R_D)(I_i - g_m V_i)$$

with

$$I_i = \frac{V_i - V_o}{R_F} = \frac{V_i - (r_d \| R_D)(I_i - g_m V_i)}{R_F}$$

and

$$I_i R_F = V_i - (r_d \| R_D) I_i + (r_d \| R_D) g_m V_i$$

so that

$$V_i[1 + g_m(r_d \| R_D)] = I_i[R_F + r_d \| R_D]$$

and finally,

$$\boxed{Z_i = \frac{V_i}{I_i} = \frac{R_F + r_d \| R_D}{1 + g_m(r_d \| R_D)}} \qquad (9.46)$$

Typically, $R_F \gg r_d \| R_D$, so that

$$Z_i \cong \frac{R_F}{1 + g_m(r_d \| R_D)}$$

For $r_d \geq 10R_D$,

$$\boxed{Z_i \cong \frac{R_F}{1 + g_m R_D}} \quad R_F \gg r_d \| R_D,\ r_d \geq 10R_D \tag{9.47}$$

Z_o: Substituting $V_i = 0$ V will result in $V_{gs} = 0$ V and $g_m V_{gs} = 0$, with a short-circuit path from gate to ground as shown in Fig. 9.39. R_F, r_d, and R_D are then in parallel and

$$\boxed{Z_o = R_F \| r_d \| R_D} \tag{9.48}$$

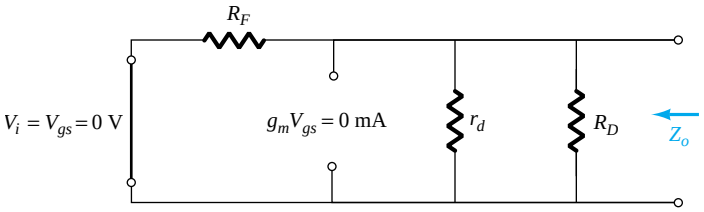


Figure 9.39 Determining Z_o for the network of Fig. 9.37.

Normally, R_F is so much larger than $r_d \| R_D$ that

$$Z_o \cong r_d \| R_D$$

and with $r_d \geq 10R_D$,

$$\boxed{Z_o \cong R_D} \quad R_F \gg r_d \| R_D,\ r_d \geq 10R_D \tag{9.49}$$

A_v: Applying Kirchhoff's current law at node D of Fig. 9.38 will result in

$$I_i = g_m V_{gs} + \frac{V_o}{r_d \| R_D}$$

but

$$V_{gs} = V_i \text{ and } I_i = \frac{V_i - V_o}{R_F}$$

so that

$$\frac{V_i - V_o}{R_F} = g_m V_i + \frac{V_o}{r_d \| R_D}$$

and

$$\frac{V_i}{R_F} - \frac{V_o}{R_F} = g_m V_i + \frac{V_o}{r_d \| R_D}$$

so that

$$V_o\left[\frac{1}{r_d \| R_D} + \frac{1}{R_F}\right] = V_i\left[\frac{1}{R_F} - g_m\right]$$

and

$$A_v = \frac{V_o}{V_i} = \frac{\left[\frac{1}{R_F} - g_m\right]}{\left[\frac{1}{r_d \| R_D} + \frac{1}{R_F}\right]}$$

but $$\frac{1}{r_d \| R_D} + \frac{1}{R_F} = \frac{1}{R_F \| r_d \| R_D}$$

and $$g_m \gg \frac{1}{R_F}$$

so that $$A_v = -g_m(R_F \| r_d \| R_D) \tag{9.50}$$

Since R_F is usually $>> r_d \| R_D$ and if $r_d \geq 10R_D$,

$$A_v \cong -g_m R_D \quad R_F \gg r_d \| R_D,\ r_d \geq 10R_D \tag{9.51}$$

Phase Relationship: The negative sign for A_v reveals that V_o and V_i are out of phase by 180°.

EXAMPLE 9.12

The E-MOSFET of Fig. 9.40 was analyzed in Example 6.11, with the result that $k = 0.24 \times 10^{-3}$ A/V^2, $V_{GS_Q} = 6.4$ V, and $I_{D_Q} = 2.75$ mA.

(a) Determine g_m.
(b) Find r_d.
(c) Calculate Z_i with and without r_d. Compare results.
(d) Find Z_o with and without r_d. Compare results.
(e) Find A_v with and without r_d. Compare results.

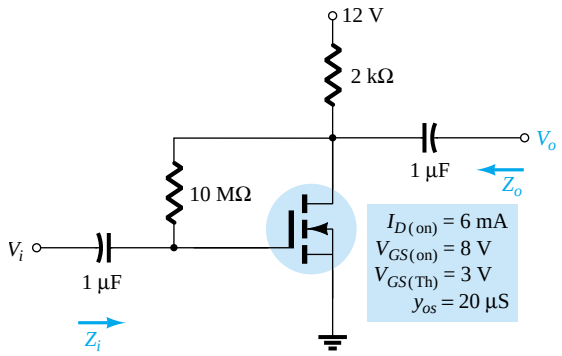


Figure 9.40 Drain-feedback amplifier from Example 6.11.

Solution

(a) $g_m = 2k(V_{GS_Q} - V_{GS(\text{Th})}) = 2(0.24 \times 10^{-3}\ \text{A/V}^2)(6.4\ \text{V} - 3\ \text{V})$
$= \mathbf{1.63\ mS}$

(b) $r_d = \dfrac{1}{y_{os}} = \dfrac{1}{20\ \mu\text{S}} = \mathbf{50\ k\Omega}$

(c) With r_d:

$$Z_i = \frac{R_F + r_d \| R_D}{1 + g_m(r_d \| R_D)} = \frac{10\ \text{M}\Omega + 50\ \text{k}\Omega \| 2\ \text{k}\Omega}{1 + (1.63\ \text{mS})(50\ \text{k}\Omega \| 2\ \text{k}\Omega)}$$
$$= \frac{10\ \text{M}\Omega + 1.92\ \text{k}\Omega}{1 + 3.13} = \mathbf{2.42\ M\Omega}$$

Without r_d:

$$Z_i \cong \frac{R_F}{1 + g_m R_D} = \frac{10\ \text{M}\Omega}{1 + (1.63\ \text{mS})(2\ \text{k}\Omega)} = \mathbf{2.53\ M\Omega}$$

revealing that since the condition $r_d \geq 10R_D = 50\ \text{k}\Omega \geq 40\ \text{k}\Omega$ is satisfied, the results for Z_o with or without r_d will be quite close.

(d) With r_d:

$$Z_o = R_F \| r_d \| R_D = 10\ \text{M}\Omega \| 50\ \text{k}\Omega \| 2\ \text{k}\Omega = 49.75\ \text{k}\Omega \| 2\ \text{k}\Omega$$
$$= \mathbf{1.92\ k\Omega}$$

Without r_d:

$$Z_o \cong R_D = \mathbf{2\ k\Omega}$$

again providing very close results.

(e) With r_d:

$$A_v = -g_m(R_F \| r_d \| R_D)$$
$$= -(1.63\ \text{mS})(10\ \text{M}\Omega \| 50\ \text{k}\Omega \| 2\ \text{k}\Omega)$$
$$= -(1.63\ \text{mS})(1.92\ \text{k}\Omega)$$
$$= \mathbf{-3.21}$$

Without r_d:

$$A_v = -g_m R_D = -(1.63\ \text{mS})(2\ \text{k}\Omega)$$
$$= \mathbf{-3.26}$$

which is very close to the above result.

9.11 E-MOSFET VOLTAGE-DIVIDER CONFIGURATION

The last E-MOSFET configuration to be examined in detail is the voltage-divider network of Fig. 9.41. The format is exactly the same as appearing in a number of earlier discussions.

Substituting the ac equivalent network for the E-MOSFET will result in the configuration of Fig. 9.42, which is exactly the same as Fig. 9.23. The result is that Eqs. (9.28) through (9.32) are applicable as listed below for the E-MOSFET.

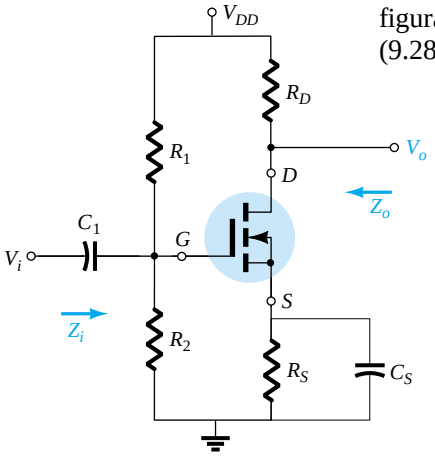


Figure 9.41 E-MOSFET voltage-divider configuration.

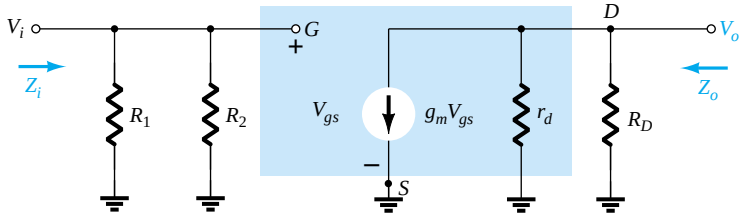


Figure 9.42 AC equivalent network for the configuration of Fig. 9.41.

Z_i:

$$Z_i = R_1 \| R_2 \tag{9.52}$$

Z_o:

$$Z_o = r_d \| R_D \tag{9.53}$$

For $r_d \geq 10R_D$,

$$Z_o \cong R_D \Big|_{r_d \geq 10R_D} \tag{9.54}$$

A_v:

$$A_v = \frac{V_o}{V_i} = -g_m(r_D \| R_D) \tag{9.55}$$

and if $r_d \geq 10R_D$,

$$A_v = \frac{V_o}{V_i} \cong -g_m R_D \tag{9.56}$$

9.12 DESIGNING FET AMPLIFIER NETWORKS

Design problems at this stage are limited to obtaining a desired dc bias condition or ac voltage gain. In most cases, the various equations developed are used "in reverse" to define the parameters necessary to obtain the desired gain, input impedance, or output impedance. To avoid unnecessary complexity during the initial stages of the design, the approximate equations are often employed because some variation will occur when calculated resistors are replaced by standard values. Once the initial design is completed, the results can be tested and refinements made using the complete equations.

Throughout the design procedure be aware that although superposition permits a separate analysis and design of the network from a dc and an ac viewpoint, a parameter chosen in the dc environment will often play an important role in the ac response. In particular, recall that the resistance R_G could be replaced by a short-circuit equivalent in the feedback configuration because $I_G \cong 0$ A for dc conditions, but for the ac analysis, it presents an important high impedance path between V_o and V_i. In addition, recall that g_m is larger for operating points closer to the I_D axis ($V_{GS} = 0$ V), requiring that R_S be relatively small. In the unbypassed R_S network, a small R_S will also contribute to a higher gain, but for the source-follower, the gain is reduced from its maximum value of 1. In total, simply keep in mind that network parameters can affect the dc and ac levels in different ways. Often a balance must be made between a particular operating point and its impact on the ac response.

In most situations, the available dc supply voltage is known, the FET to be employed has been determined, and the capacitors to be employed at the chosen frequency are defined. It is then necessary to determine the resistive elements necessary to establish the desired gain or impedance level. The next three examples will determine the required parameters for a specific gain.

EXAMPLE 9.13

Design the fixed-bias network of Fig. 9.43 to have an ac gain of 10. That is, determine the value of R_D.

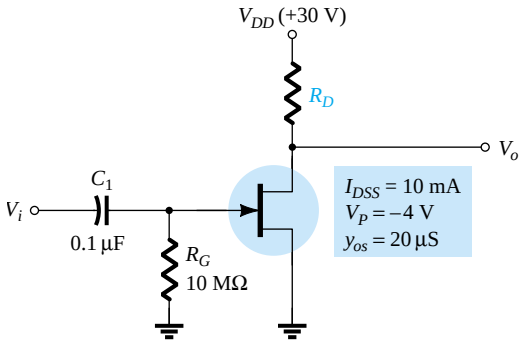


Figure 9.43 Circuit for desired voltage gain in Example 9.13.

Solution

Since $V_{GS_Q} = 0$ V, the level of g_m is g_{m0}. The gain is therefore determined by

$$A_v = -g_m(R_D \| r_d) = -g_{m0}(R_D \| r_d)$$

with

$$g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(10\text{ mA})}{4\text{ V}} = 5\text{ mS}$$

The result is

$$-10 = -5\text{ mS}(R_D \| r_d)$$

and

$$R_D \| r_d = \frac{10}{5\text{ mS}} = 2\text{ k}\Omega$$

From the device specifications,

$$r_d = \frac{1}{y_{os}} = \frac{1}{20 \times 10^{-6}\text{ S}} = 50\text{ k}\Omega$$

Substituting, we find

$$R_D \| r_d = R_D \| 50\text{ k}\Omega = 2\text{ k}\Omega$$

and

$$\frac{R_D(50\text{ k}\Omega)}{R_D + 50\text{ k}\Omega} = 2\text{ k}\Omega$$

or

$$50R_D = 2(R_D + 50\text{ k}\Omega) = 2R_D + 100\text{ k}\Omega$$

with

$$48R_D = 100\text{ k}\Omega$$

and

$$R_D = \frac{100\text{ k}\Omega}{48} \cong 2.08\text{ k}\Omega$$

The closest standard value is **2 kΩ** (Appendix C), which would be employed for this design.

The resulting level of V_{DS_Q} would then be determined as follows:

$$V_{DS_Q} = V_{DD} - I_{D_Q}R_D = 30\text{ V} - (10\text{ mA})(2\text{ k}\Omega) = \mathbf{10\text{ V}}$$

The levels of Z_i and Z_o are set by the levels of R_G and R_D, respectively. That is,

$$Z_i = R_G = \mathbf{10\text{ M}\Omega}$$

$$Z_o = R_D \| r_d = 2\text{ k}\Omega \| 50\text{ k}\Omega = \mathbf{1.92\text{ k}\Omega} \cong R_D = 2\text{ k}\Omega.$$

EXAMPLE 9.14

Choose the values of R_D and R_S for the network of Fig. 9.44 that will result in a gain of 8 using a relatively high level of g_m for this device defined at $V_{GS_Q} = \frac{1}{4}V_P$.

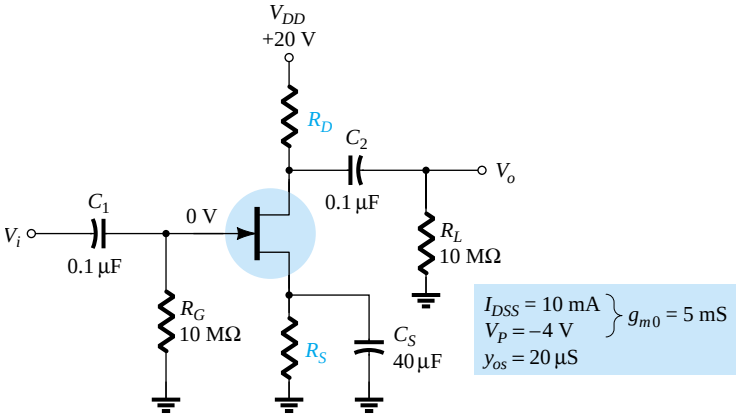


Figure 9.44 Network for desired voltage gain in Example 9.14.

Solution

The operating point is defined by

$$V_{GS_Q} = \frac{1}{4}V_P = \frac{1}{4}(-4\text{ V}) = -1\text{ V}$$

and

$$I_D = I_{DSS}\left(1 - \frac{V_{GS_Q}}{V_P}\right)^2 = 10\text{ mA}\left(1 - \frac{(-1\text{ V})}{(-4\text{ V})}\right)^2 = 5.625\text{ mA}$$

Determining g_m,

$$g_m = g_{m0}\left(1 - \frac{V_{GS_Q}}{V_P}\right)$$

$$= 5\text{ mS}\left(1 - \frac{(-1\text{ V})}{(-4\text{ V})}\right) = 3.75\text{ mS}$$

The magnitude of the ac voltage gain is determined by

$$|A_v| = g_m(R_D \| r_d)$$

Substituting known values will result in

$$8 = (3.75\text{ mS})(R_D \| r_d)$$

so that

$$R_D \| r_d = \frac{8}{3.75\text{ mS}} = 2.13\text{ k}\Omega$$

The level of r_d is defined by

$$r_d = \frac{1}{y_{os}} = \frac{1}{20\ \mu\text{S}} = 50\text{ k}\Omega$$

and

$$R_D \| 50\text{ k}\Omega = 2.13\text{ k}\Omega$$

with the result that

$$R_D = \mathbf{2.2\ k\Omega}$$

which is a standard value.

g_m

The level of R_S is determined by the dc operating conditions as follows:

$$V_{GS_Q} = -I_D R_S$$

$$-1\text{ V} = -(5.625\text{ mA})R_S$$

and

$$R_S = \frac{1\text{ V}}{5.625\text{ mA}} = 177.8\ \Omega$$

The closest standard value is 180 Ω. In this example, R_S does not appear in the ac design because of the shorting effect of C_S.

In the next example, R_S is unbypassed and the design becomes a bit more complicated.

EXAMPLE 9.15

Determine R_D and R_S for the network of Fig. 9.44 to establish a gain of 8 if the bypass capacitor C_S is removed.

Solution

V_{GS_Q} and I_{D_Q} are still −1 V and 5.625 mA, and since the equation $V_{GS} = -I_D R_S$ has not changed, R_S continues to equal the standard value of **180 Ω** obtained in Example 9.14.

The gain of an unbypassed self-bias configuration is

$$A_v = -\frac{g_m R_D}{1 + g_m R_S}$$

For the moment it is assumed that $r_d \geq 10(R_D + R_S)$. Using the full equation for A_v at this stage of the design would simply complicate the process unnecessarily.

Substituting (for the specified magnitude of 8 for the gain),

$$|8| = \left|\frac{-(3.75\text{ mS})R_D}{1 + (3.75\text{ mS})(180\ \Omega)}\right| = \frac{(3.75\text{ mS})R_D}{1 + 0.675}$$

and

$$8(1 + 0.675) = (3.75\text{ mS})R_D$$

so that

$$R_D = \frac{13.4}{3.75\text{ mS}} = 3.573\ \text{k}\Omega$$

with the closest standard value at **3.6 kΩ.**

We can now test the condition:

$$r_d \geq 10(R_D + R_S)$$

$$50\ \text{k}\Omega \geq 10(3.6\ \text{k}\Omega + 0.18\ \text{k}\Omega) = 10(3.78\ \text{k}\Omega)$$

and

$$50\ \text{k}\Omega \geq 37.8\ \text{k}\Omega$$

which is satisfied—the solution stands!

9.13 SUMMARY TABLE

In an effort to provide a quick comparison between configurations and offer a listing that can be helpful for a variety of reasons, Table 9.1 was developed. The exact and approximate equation for each important parameter are provided with a typical range of values for each. Although all the possible configurations are not present, the majority of the most frequently encountered are included. In fact, any configuration not

TABLE 9.1 Z_i, Z_0, and A_v for various FET configurations

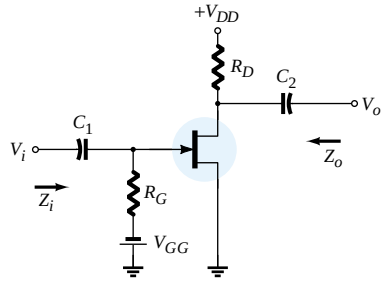
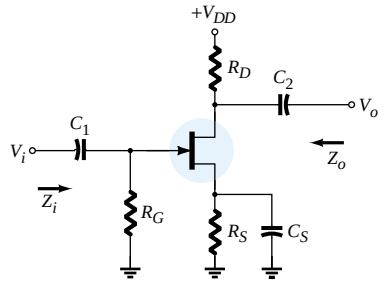
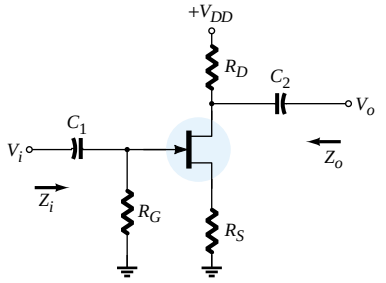
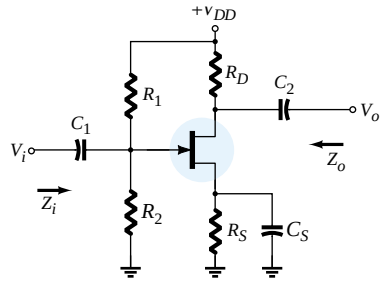
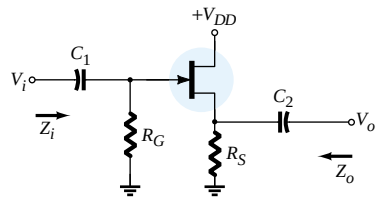
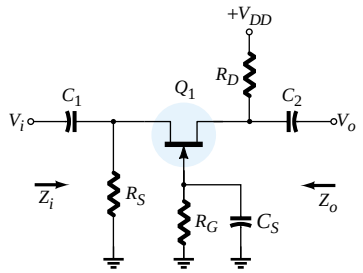
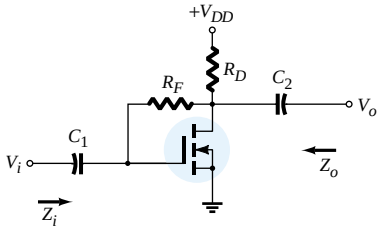
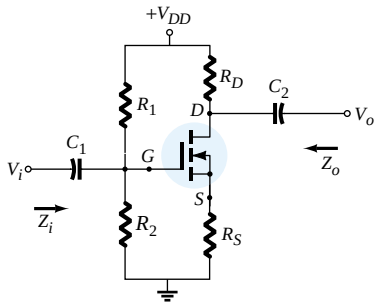
Configuration	Z_i	Z_o	$A_v = \frac{V_o}{V_i}$
Fixed-bias [JFET or D-MOSFET]	 High (10 MΩ) $= R_G$	Medium (2 kΩ) $= R_D \| r_d$ $\cong R_D$ $(r_d \geq 10\,R_D)$	Medium (−10) $= -g_m(r_d \| R_D)$ $\cong -g_m R_D$ $(r_d \geq 10\,R_D)$
Self-bias bypassed R_s [JFET or D-MOSFET]	 High (10 MΩ) $= R_G$	Medium (2 kΩ) $= R_D \| r_d$ $\cong R_D$ $(r_d \geq 10\,R_D)$	Medium (−10) $= -g_m(r_d \| R_D)$ $\cong -g_m R_D$ $(r_d \geq 10\,R_D)$
Self-bias unbypassed R_S [JFET or D-MOSFET]	 High (10 MΩ) $= R_G$	$= \dfrac{\left[1 + g_m R_S + \frac{R_S}{r_d}\right] R_D}{\left[1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d}\right]}$ $= R_D$ $r_d \geq 10\,R_D$ or $r_d = \infty\Omega$	Low (−2) $= \dfrac{g_m R_D}{1 + g_m R_s + \frac{R_D + R_S}{r_d}}$ $\cong -\dfrac{g_m R_D}{1 + g_m R_S}$ $[r_d \geq 10(R_d + R_S)]$
Voltage-divider bias [JFET or D-MOSFET]	 High (10 MΩ) $= R_1 \| R_2$	Medium (2 kΩ) $= R_D \| r$ $\cong R_D$ $(r_d \geq 10\,R_D)$	Medium (−10) $= -g_m(r_d \| R_D)$ $\cong -g_m R_D$ $(r_d \geq 10\,R_D)$

TABLE 9.1 (Continued)

Configuration	Z_i	Z_o	$A_v = \frac{V_o}{V_i}$
Source-follower [JFET or D-MOSFET] 	High (10 MΩ) $= R_G$	Low (100 kΩ) $= r_d \| R_S \| 1/g_m$ $\cong R_S \| 1/g_m \quad (r_d \geq 10\,R_S)$	Low (< 1) $= \frac{g_m(r_d \| R_S)}{1 + g_m(r_d \| R_S)}$ $\cong \frac{g_m R_S}{1 + g_m R_S} \quad (r_d \geq 10\,R_S)$
Common-gate [JFET or D-MOSFET] 	Low (1 kΩ) $= R_S \left\| \left[\frac{r_d + R_D}{1 + g_m r_d} \right] \right.$ $\cong R_S \| \frac{1}{g_m} \quad (r_d \geq 10\,R_D)$	Medium (2 kΩ) $= R_D \| r_d$ $\cong R_D \quad (r_d \geq 10\,R_D)$	Medium (+10) $= \frac{g_m R_D + \frac{R_D}{r_d}}{1 + \frac{R_D}{r_d}}$ $\cong g_m R_D \quad (r_d \geq 10\,R_S)$
Drain-feedback bias E-MOSFET 	Medium (1 MΩ) $= \frac{R_F + r_d \| R_D}{1 + g_m(r_d \| R_D)}$ $\cong \frac{R_F}{1 + g_m R_D} \quad (r_d \geq 10\,R_D)$	Medium (2 kΩ) $= R_F \| r_d \| R_D$ $\cong R_D \quad (R_F, r_d \geq 10\,R_D)$	Medium (−10) $= -g_m(R_F \| r_d \| R_D)$ $\cong -g_m R_D \quad (R_F, r_d \geq 10\,R_D)$
Voltage-divider bias E-MOSFET 	Medium (1 MΩ) $= R_1 \| R_2$	Medium (2 kΩ) $= R_D \| r_d$ $\cong R_D \quad (R_d \geq 10\,R_D)$	Medium (−10) $= -g_m(r_d \| R_D)$ $\cong -g_m R_D \quad (r_d \geq 10\,R_D)$

listed will probably be some variation of those appearing in the table, so at the very least, the listing will provide some insight as to what expected levels should be and which path will probably generate the desired equations. The format chosen was designed to permit a duplication of the entire table on the front and back of one $8\frac{1}{2}$ by 11 inch page.

9.14 TROUBLESHOOTING

As mentioned before, troubleshooting a circuit is a combination of knowing the theory and having experience using meters and an oscilloscope to check the operation of the circuit. A good troubleshooter has a "nose" for finding the trouble in a circuit—this ability to "see" what is happening being greatly developed through building, testing, and repairing many different circuits. For an FET small-signal amplifier, one could go about troubleshooting a circuit by performing a number of basic steps:

1. Look at the circuit board to see if any obvious problems can be seen: an area charred by excess heating of a component; a component that feels or seems too hot to touch; what appears to be a poor solder joint; any connection that appears to have come loose.
2. Use a dc meter: make some measurements as marked in a repair manual containing the circuit schematic diagram and a listing of test dc voltages.
3. Apply a test ac signal: measure the ac voltages starting at the input and working along toward the output.
4. If the problem is identified at a particular stage, the ac signal at various points should be checked using an oscilloscope to see the waveform, its polarity, amplitude, and frequency, as well as any unusual waveform "glitches" that may be present. In particular, observe that the signal is present for the full signal cycle.

Possible Symptoms and Actions

If there is no output ac voltage:

1. Check if the supply voltage is present.
2. Check if the output voltage at V_D is between 0 V and V_{DD}.
3. Check if there is any input ac signal at the gate terminal.
4. Check the ac voltage at each side of the coupling capacitor terminals.

When building and testing a FET amplifier circuit in the laboratory:

1. Check the color code of resistor values to be sure that they are correct. Even better, measure the resistor value as components used repeatedly may get overheated when used incorrectly, causing the nominal value to change.
2. Check that all dc voltages are present at the component terminals. Be sure that all ground connections are made common.
3. Measure the ac input signal to be sure the expected value is provided to the circuit.

9.15 PSPICE WINDOWS

JFET Fixed-Bias Configuration

The first JFET configuration to be analyzed using PSpice Windows is the fixed-bias configuration of Fig. 9.45, which has a JFET with $V_P = -4$ V and $I_{DSS} =$ 10 mA. The 10-MΩ resistor was added to act as a path to ground for the capacitor

but is essentially an open-circuit as a load. The **J2N3819 *n*-channel JFET** from the **EVAL.slb** library will be used, and the ac voltage will be determined at four different points for comparison and review.

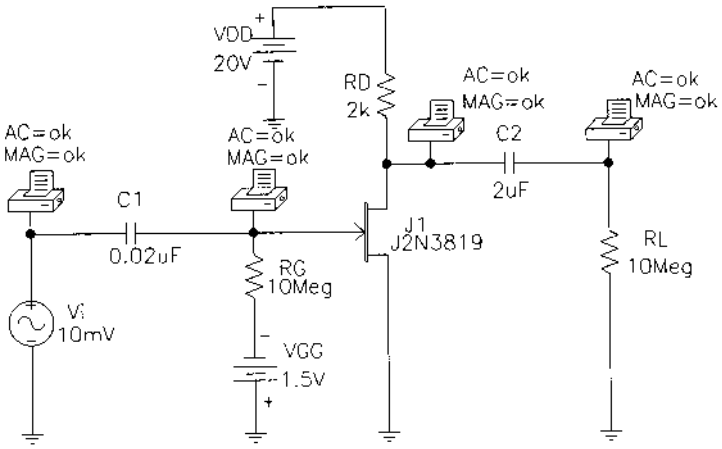


Figure 9.45 Fixed-bias JFET configuration with an ac source.

The constant **Beta** is determined by

$$\text{Beta} = \frac{I_{DSS}}{|V_p|^2} = \frac{10\text{ mA}}{4^2} = 0.625\text{ mA/V}^2$$

and inserted as a **Model Parameter** using the sequence **Edit-Model-Edit Instance Model (Text). Vto** must also be changed to -4 V. The remaining elements of the network are set as described for the transistor in Chapter 8.

An analysis of the network will result in the printout of Fig. 9.46. The **Schematics Netlist** reveals the nodes assigned to each parameter and defines the nodes for which the ac voltage is to be printed. In this case, note that **Vi** is set at 10 mV at a frequency of 10kHz from node 2 to 0. In the list of **Junction FET MODEL PARAMETERS, VTO** is -4 V and **BETA** is 625E-6 as entered. The **SMALL-SIGNAL BIAS SOLUTION** reveals that the voltage at both ends of R_G is -1.5 V, resulting in $V_{GS} = -1.5$ V. The voltage from drain to source (ground) is 12 V, leaving a drop of 8 V across R_D. The **AC ANALYSIS** at the end of the listing reveals that the voltage at the source (node 2) is 10 mV as set, but the voltage at the other end of the capacitor is 3 μV less due to the impedance of the capacitor at 10 kHz—certainly a drop to be ignored. The choice of 0.02 μ F for this frequency was obviously a good one. The voltages before and after the capacitor on the output side are exactly the same (to three places), revealing that the larger the capacitor, the closer the characteristics to a short circuit. The output of 6.275E-2 = 62.75 mV reflects a gain of 6.275. The **OPERATING POINT INFORMATION** reveals that I_D is 4 mA and g_m is 3.2 mS. Calculating the value of g_m from:

$$g_m = \frac{2I_{DSS}}{|V_P|}\left(1 - \frac{V_{GS_Q}}{V_P}\right)$$
$$= \frac{2\ (10\text{ mA})}{4\text{ V}}\left(1 - \frac{(-1.5\text{ V})}{(-4\text{ V})}\right)$$
$$= 3.125\text{ mS}$$

confirming our analysis.

```
 ****     CIRCUIT DESCRIPTION
******************************************************************************
* Schematics Netlist *

V_VDD          $N_0001 0 20V
C_C1           $N_0002 $N_0003  0.02uF
R_RG           $N_0004 $N_0003  10Meg
R_RD           $N_0005 $N_0001  2k
V_VGG          0 $N_0004 1.5V

.PRINT          AC
+ VM([$N_0003])

.PRINT          AC
+ VM([$N_0002])

.PRINT          AC
+ VM([$N_0005])

.PRINT          AC
+ VM([$N_0006])
C_C2           $N_0005 $N_0006  2uF
V_Vi           $N_0002 0 AC 10mV
+SIN 0 10mV 10kHz 0 0 0
R_RL           0 $N_0006  10Meg
J_J1           $N_0005 $N_0003 0 J2N3819-X

 ****     Junction FET MODEL PARAMETERS
******************************************************************************
               J2N3819-X
               NJF
         VTO   -4
        BETA  625.000000E-06
      LAMBDA    2.250000E-03
          IS   33.570000E-15
         ISR  322.400000E-15
       ALPHA  311.700000E-06
          VK  243.6
          RD    1
          RS    1
         CGD    1.600000E-12
         CGS    2.414000E-12
           M     .3622
       VTOTC   -2.500000E-03
     BETATCE    -.5
          KF    9.882000E-18

 ****     SMALL SIGNAL BIAS SOLUTION       TEMPERATURE =   27.000 DEG C
******************************************************************************
 NODE   VOLTAGE     NODE   VOLTAGE     NODE   VOLTAGE     NODE   VOLTAGE

($N_0001)   20.0000                  ($N_0002)    0.0000

($N_0003)   -1.5000                  ($N_0004)   -1.5000

($N_0005)   12.0020                  ($N_0006)    0.0000

    VOLTAGE SOURCE CURRENTS
    NAME         CURRENT

    V_VDD       -3.999E-03
    V_VGG       -1.366E-12

 ****     OPERATING POINT INFORMATION      TEMPERATURE =   27.000 DEG C
******************************************************************************
**** JFETS

NAME         J_J1
MODEL        J2N3819-X
ID           4.00E-03
VGS         -1.50E+00
VDS          1.20E+01
GM           3.20E-03
GDS          8.76E-06
CGS          1.73E-12
CGD          6.07E-13 .
**** 10/08/97 11:23:59 ******** NT Evaluation PSpice (October 1996) **********

 ****     AC ANALYSIS                      TEMPERATURE =   27.000 DEG C
******************************************************************************
  FREQ        VM($N_0003)

   1.000E+04   9.997E-03


  FREQ        VM($N_0002)

   1.000E+04   1.000E-02


  FREQ        VM($N_0005)

   1.000E+04   6.275E-02


  FREQ        VM($N_0006)

   1.000E+04   6.275E-02
```

Figure 9.46 Output file for the network of Figure 9.45.

JFET Self-Bias Configuration

The self-bias configuration of Fig. 9.47 will be analyzed using the **J2N3819 JFET** from the library and then using an approximate equivalent circuit. It will be interesting to see if there are any major differences in solution.

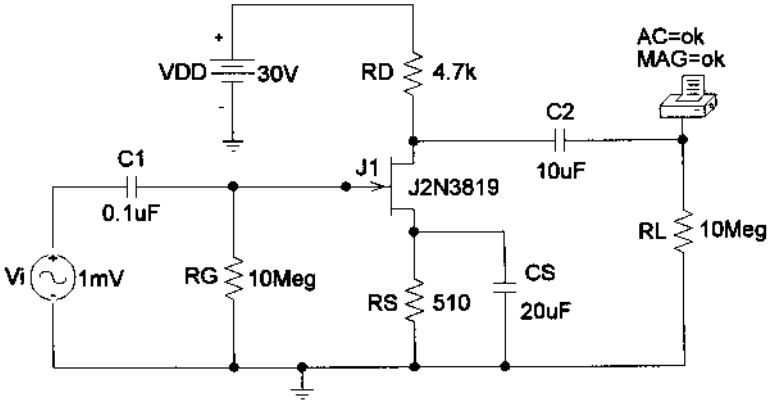


Figure 9.47 Self-bias configuration with an ac source.

Again, $V_P = -4$ V and $I_{DSS} = 10$ mA, resulting in a **Vto** of -4 and a **Beta** of 6.25E-4. The **Analysis** is run and the results of Fig. 9.48 obtained. The nodes are identified in the **Schematics Netlist** and the parameters in the **Junction FET MODEL PARAMETERS.** The **SMALL-SIGNAL BIAS SOLUTION** reveals that $V_{GS} = -1.7114$ V and $V_D = 14.228$ V—results that are very close to a hand-written solution of -1.68 V and 14.49 V. The **OPERATING POINT INFORMATION** reveals that I_D is 3.36 mA compared to a hand-calculated level of 3.3 mA and that g_m is 2.94 mS compared to a hand-calculated level of 2.90 mS. The **AC ANALYSIS** provides an output level of 13.3 mV at an angle of $-179.9°$, which compares well with a hand-calculated level of 13.63 mV at an angle of $-180°$. The results for JFETs are a lot closer than those obtained for transistors when we used the provided elements because of the special feature of having essentially infinite input impedance so that the gate current is zero ampere. Recall that for the transistor, V_{BE} is a function of the operating conditions.

We will now investigate the self-bias configuration using the approximate model as done for the transistor and see if there is an improvement in the results (compared to the hand-calculated levels). In this case, we need the voltage controlled current source **(VCCS)** found in the **ANALOG.slb** library as **G.** When selected, the **Description** reads **Voltage-controlled current source.** When placed on the schematic, it will appear as shown in Fig. 9.49. The sensing voltage is between the plus and minus sign, while the controlled current is between the other two external terminals.

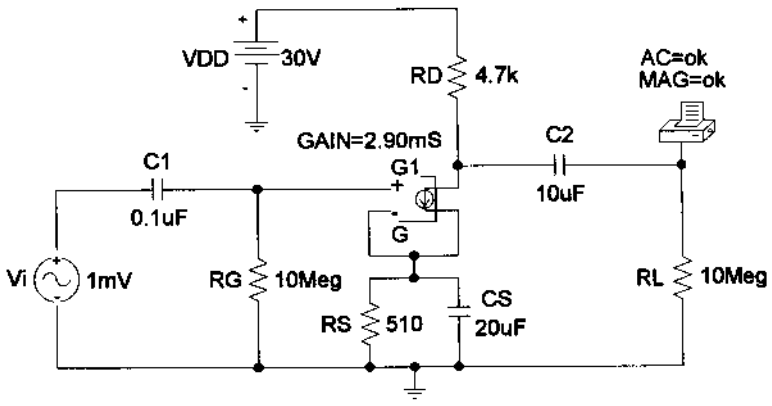


Figure 9.49 Network in Figure 9.47 following substitution of a **VCCS** for the JFET in the ac domain.

```
 ****     CIRCUIT DESCRIPTION
******************************************************************************
* Schematics Netlist *


V_Vi          $N_0001 0  AC 1mV
+SIN 0V 1mV 10kHz 0 0 0
C_C1          $N_0001 $N_0002  0.1uF
R_RG          $N_0002 0  10Meg
V_VDD          $N_0003 0 30V
R_RD          $N_0003 $N_0004  4.7k
R_RS          $N_0005 0  510
C_CS          $N_0005 0  20uF
C_C2          $N_0004 $N_0006  10uF
R_RL          $N_0006 0  10Meg
J_J1          $N_0004 $N_0002 $N_0005 J2N3819-X

.PRINT          AC
+ VM([$N_0006])
+ VP([$N_0006])

 ****     Junction FET MODEL PARAMETERS

******************************************************************************

               J2N3819-X
               NJF
        VTO     -4
       BETA   625.000000E-06
     LAMBDA     2.250000E-03
         IS    33.570000E-15
        ISR   322.400000E-15
      ALPHA   311.700000E-06
         VK   243.6
         RD     1
         RS     1
        CGD     1.600000E-12
        CGS     2.414000E-12
          M      .3622
      VTOTC    -2.500000E-03
    BETATCE     -.5
         KF     9.882000E-18

 ****     SMALL SIGNAL BIAS SOLUTION       TEMPERATURE =   27.000 DEG C

******************************************************************************

 NODE   VOLTAGE     NODE   VOLTAGE     NODE   VOLTAGE     NODE   VOLTAGE

($N_0001)    0.0000                   ($N_0002) 13.95E-06

($N_0003)   30.0000                   ($N_0004)   14.2280

($N_0005)    1.7114                   ($N_0006)    0.0000


    VOLTAGE SOURCE CURRENTS
    NAME         CURRENT

    V_Vi          0.000E+00
    V_VDD        -3.356E-03

    TOTAL POWER DISSIPATION   1.01E-01  WATTS

 ****     OPERATING POINT INFORMATION      TEMPERATURE =   27.000 DEG C

******************************************************************************

**** JFETS


NAME          J_J1
MODEL         J2N3819-X
ID            3.36E-03
VGS          -1.71E+00
VDS           1.25E+01
GM            2.94E-03
GDS           7.34E-06
CGS           1.68E-12
CGD           5.97E-13

 ****     AC ANALYSIS                      TEMPERATURE =   27.000 DEG C

******************************************************************************

  FREQ        VM($N_0006) VP($N_0006)

   1.000E+04   1.330E-02  -1.799E+02
```

Figure 9.48 Output file for the network of Figure 9.47.

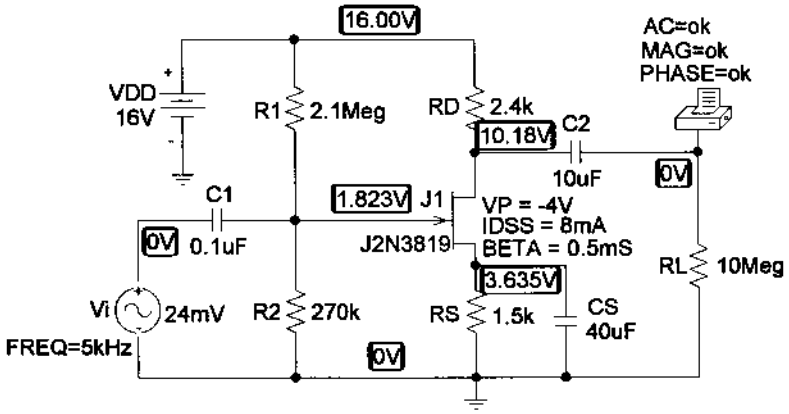


Figure 9.50 JFET voltage-divider configuration with an ac source.

Double-clicking on the schematic symbol will result in a **PartName: G** dialog box, in which the **GAIN(g_m)** can be set to the hand-calculated level of 2.90 mS.

The result of an analysis is a gain of 13.62—almost an exact match with the hand-written gain. This approach is certainly valid for an ac analysis, but if we examine the **SMALL-SIGNAL BIAS SOLUTION,** we find that the results are meaningless. Therefore, the equivalent appearing in Fig. 9.49 is only valid for the ac gain since the only parameter defined is the ac transconductance factor.

JFET Voltage-Divider Configuration

The last network to be analyzed in this PSpice Windows presentation is the voltage-divider configuration of Fig. 9.50. Note that the parameters chosen are different from those employed in earlier examples, with V_i at 24 mV and a frequency of 5 kHz. In addition, the dc levels are displayed and a plot of the output and input voltages will be obtained on the same screen.

After setting up the network, the source V_i must be set to the indicated parameters by double-clicking on the source and then sequentially double-clicking on each parameter and typing in the correct values. Each must be saved and then the display changed to print the magnitude of the ac voltage and the applied frequency. In this example, the JFET parameters were printed on the screen using the **ABC** icon. **BETA** is of course calculated from $I_{DSS}/|V_P|^2$. Under **Analysis-Probe Setup,** the option **Do not auto-run Probe** was chosen, and under **Setup, AC Sweep** was chosen and the frequency of 5 kHz entered. Finally, since we want the dc levels to be displayed, the **Display Results on Schematic** option is chosen under **Analysis,** and **Enable Voltage Display** is enabled. The resulting dc levels of Fig. 9.50 reveal that V_{GS} is 1.823 V − 3.635 V = −1.812 V, comparing very well with the −1.8 V calculated in Example 6.5. V_D is 10.18 V compared to the calculated level of 10.24 V, and V_{DS} is 10.18 V − 3.635 V = 6.545 V compared to 6.64 V.

For the ac solution, we can choose **Examine Output** under **Analysis** and find under **OPERATING POINT INFORMATION** that g_m is 2.22 mS, comparing very well with the hand-calculated value of 2.2 mS, and under **AC ANALYSIS** that the output ac voltage is 125.8 mV, resulting in a gain of 125.8 mV/24 mV = 5.24. The hand-calculated level is $g_m R_D$ = (2.2 mS)(2.4 kΩ) = 5.28. The ac waveform for the output can be obtained by first applying the sequence **Analysis-Probe Setup-Automatically run Probe after simulation.** Then, return to **Setup** under **Analysis,** and enable **Transient,** disable **AC Sweep,** and double-click **Transient** to obtain the **Transient** dialog box. For the frequency of 5 kHz, the period is 200 μs. A **Print Step** to 2 μs would then give us 100 plot points for each cycle. The **Final Time** will be 5 × 200 μs = 1 ms to show five cycles. The **No-Print Delay** will be 0s and the **Step Ceiling** 2μs. Then, click the **Trace** icon, choose **V(J1:d),** and the output waveform of Fig. 9.51 will appear. Choose **Plot-Add Plot-Trace-Add-V(Vi:+),** and both wave-

forms will appear as shown. Shift **SEL≫** to the bottom waveform by simply bringing the pointer to the left of the lower waveform and left-clicking the mouse once. Click the **Toggle cursor** icon, and a horizontal line will appear at the dc level of the output voltage at 10.184 V. A left click of the mouse and an intersecting set of lines will appear. Choose the **Cursor Peak** icon, and the intersection will automatically go to the peak value of the waveform **(A1** in the dialog box). The difference appearing in the dialog box is 125.496 mV, comparing well with the printed value in the output file. The difference is simply due to the number of points chosen for the plot; an increased number of plot points would have brought the two levels closer together.

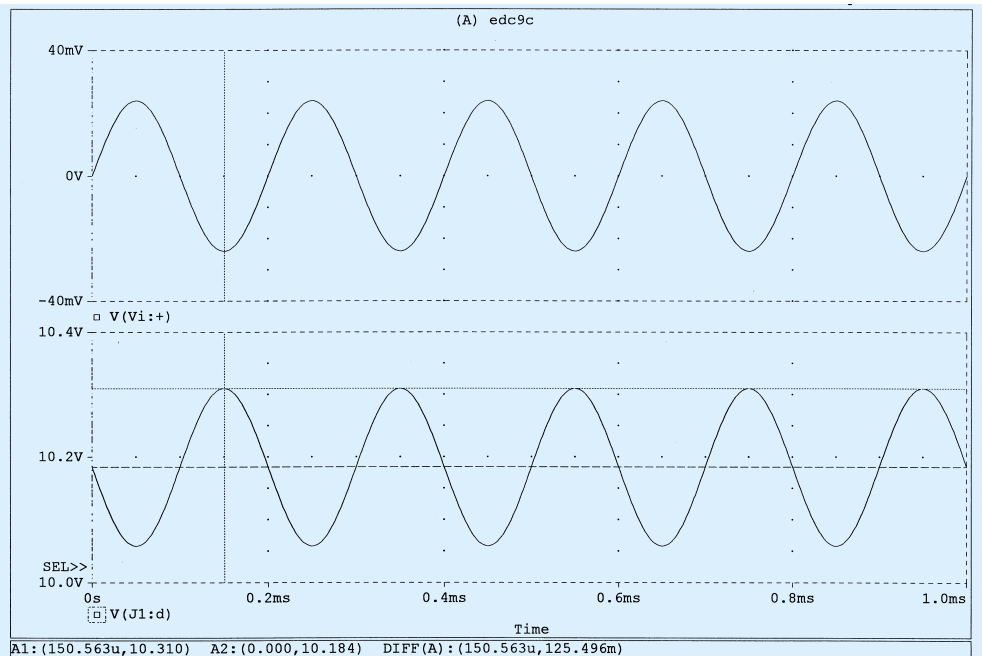


Figure 9.51 The ac drain and gate voltage for the voltage-divider JFET configuration of Figure 9.50.

§ 9.2 FET Small-Signal Model

PROBLEMS

1. Calculate g_{m0} for a JFET having device parameters $I_{DSS} = 15$ mA and $V_P = -5$ V.

2. Determine the pinch-off voltage of a JFET with $g_{m0} = 10$ mS and $I_{DSS} = 12$ mA.

3. For a JFET having device parameters $g_{m0} = 5$ mS and $V_P = -3.5$ V, what is the device current at $V_{GS} = 0$ V?

4. Calculate the value of g_m for a JFET ($I_{DSS} = 12$ mA, $V_P = -3$ V) at a bias point of $V_{GS} = -1$ V.

5. For a JFET having $g_m = 6$ mS at $V_{GS_Q} = -1$ V, what is the value of I_{DSS} if $V_P = -2.5$ V?

6. A JFET ($I_{DSS} = 10$ mA, $V_P = -5$ V) is biased at $I_D = I_{DSS}/4$. What is the value of g_m at that bias point?

7. Determine the value of g_m for a JFET ($I_{DSS} = 8$ mA, $V_P = -5$ V) when biased at $V_{GS_Q} = V_P/4$.

8. A specification sheet provides the following data (at a listed drain–source current)

$$y_{fs} = 4.5 \text{ mS}, \qquad y_{os} = 25\ \mu\text{S}$$

At the listed drain–source current, determine:
(a) g_m.
(b) r_d.

9. For a JFET having specified values of $y_{fs} = 4.5$ mS and $y_{os} = 25\ \mu$S, determine the device output impedance, Z_o(FET), and device ideal voltage gain, A_v(FET).

10. If a JFET having a specified value of $r_d = 100\ \text{k}\Omega$ has an ideal voltage gain of $A_v(\text{FET}) = -200$, what is the value of g_m?

11. Using the transfer characteristic of Fig. 9.52:
(a) What is the value of g_{m0}?
(b) Determine g_m at $V_{GS} = -1.5$ V graphically.
(c) What is the value of g_m at $V_{GS_Q} = -1.5$ V using Eq. (9.6)? Compare with the solution to part (b).
(d) Graphically determine g_m at $V_{GS} = -2.5$ V.
(e) What is the value of g_m at $V_{GS_Q} = -2.5$ V using Eq. (9.6)? Compare with the solution to part (d).

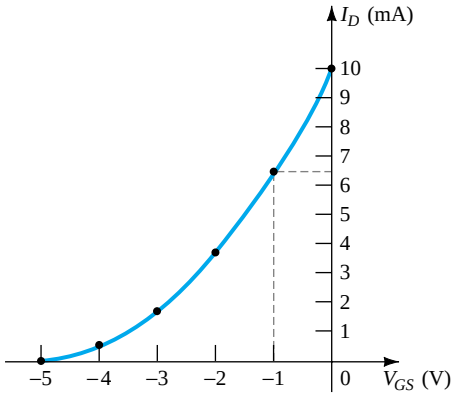


Figure 9.52 JFET transfer characteristic for Problem 11

12. Using the drain characteristic of Fig. 9.53:
(a) What is the value of r_d for $V_{GS} = 0$ V?
(b) What is the value of g_{m0} at $V_{DS} = 10$ V?

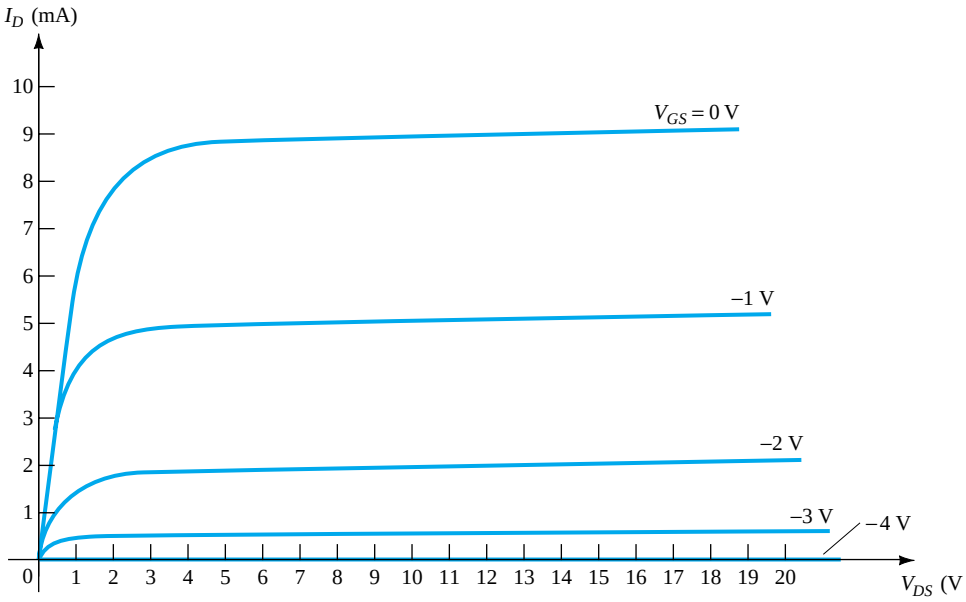


Figure 9.53 JFET drain characteristic for Problem 12

13. For a 2N4220 *n*-channel JFET (y_{fs}(minimum) = 750 μS, y_{os}(maximum) = 10 μS):
(a) What is the value of g_m?
(b) What is the value of r_d?

14. (a) Plot g_m vs. V_{GS} for an *n*-channel JFET with $I_{DSS} = 8$ mA and $V_P = -6$ V.
(b) Plot g_m vs. I_D for the same *n*-channel JFET as part (a).

15. Sketch the ac equivalent model for a JFET if $y_{fs} = 5.6$ mS and $y_{os} = 15\ \mu$S.

16. Sketch the ac equivalent model for a JFET if $I_{DSS} = 10$ mA, $V_P = -4$ V, $V_{GS_Q} = -2$ V, and $y_{os} = 25\ \mu$S.

§ 9.3 JFET Fixed-Bias Configuration

17. Determine Z_i, Z_o and A_v for the network of Fig. 9.54 if $I_{DSS} = 10$ mA, $V_P = -4$ V, and $r_d = 40$ kΩ.

18. Determine Z_i, Z_o, and A_v for the network of Fig. 9.54 if $I_{DSS} = 12$ mA, $V_P = -6$ V, and $y_{os} = 40\ \mu$S.

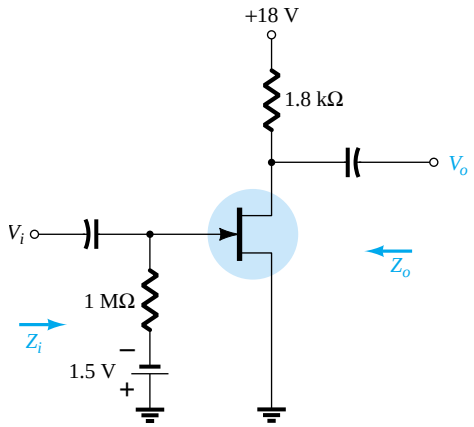


Figure 9.54 Fixed-bias amplifier for Problems 17 and 18

§ 9.4 JFET Self-Bias Configuration

19. Determine Z_i, Z_o, and A_v for the network of Fig. 9.55 if $y_{fs} = 3000\ \mu$S and $y_{os} = 50\ \mu$s.

20. Determine Z_i, Z_o, and A_v for the network of Fig. 9.56 if $I_{DSS} = 6$ mA, $V_P = -6$ V, and $y_{os} = 40\ \mu$S.

21. Determine Z_i, Z_o, and A_v for the network of Fig. 9.55 if the 20-μF capacitor is removed and the parameters of the network are the same as in Problem 19. Compare results with those of Problem 19.

22. Repeat Problem 19 if y_{os} is 10 μS. Compare the results to those of Problem 19.

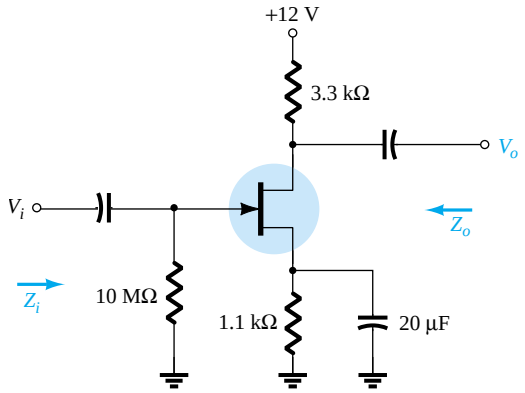


Figure 9.55 Problems 19, 21, and 46

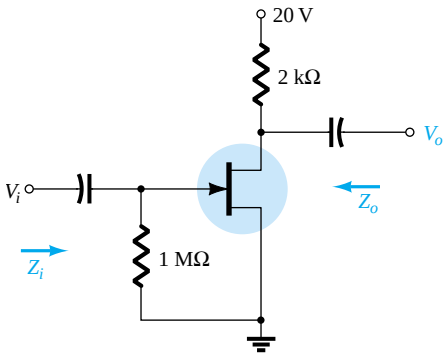


Figure 9.56 Self-bias configuration for Problems 20 and 47

§ 9.5 JFET Voltage-Divider Configuration

23. Determine Z_i, Z_o, and V_o for the network of Fig. 9.57 if $V_i = 20$ mV.

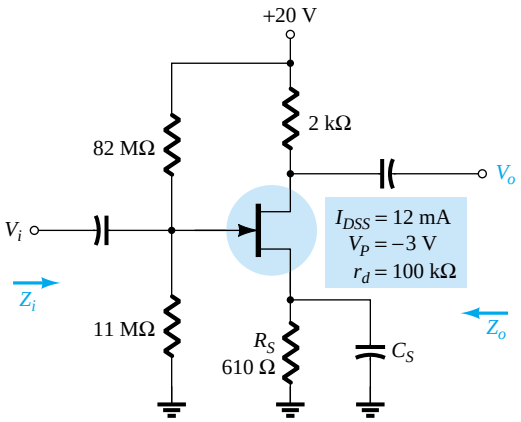


Figure 9.57 Problems 23–26 and 48

24. Determine Z_i, Z_o, and V_o for the network of Fig. 9.57 if $V_i = 20$ mV and the capacitor C_S is removed.

25. Repeat Problem 23 if $r_d = 20$ kΩ and compare results.

26. Repeat Problem 24 if $r_d = 20$ kΩ and compare results.

§ 9.6 JFET Source-Follower Configuration

27. Determine Z_i, Z_o, and A_v for the network of Fig. 9.58.

28. Repeat Problem 27 if $r_d = 20$ kΩ.

29. Determine Z_i, Z_o, and A_v for the network of Fig. 9.59.

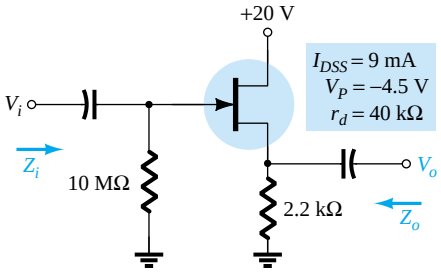


Figure 9.58 Problems 27 and 28

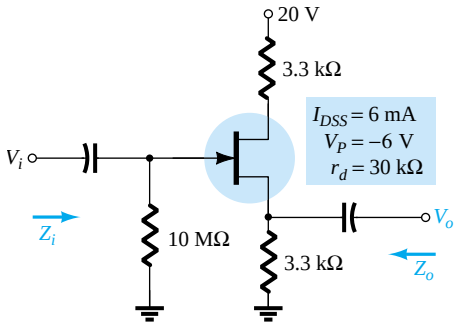


Figure 9.59 Problem 29

§ 9.7 JFET Common-Gate Configuration

30. Determine Z_i, Z_o, and V_o for the network of Fig. 9.60 if $V_i = 0.1$ mV.

31. Repeat Problem 30 if $r_d = 25$ kΩ.

32. Determine Z_i, Z_o, and A_v for the network of Fig. 9.61 if $r_d = 33$ kΩ.

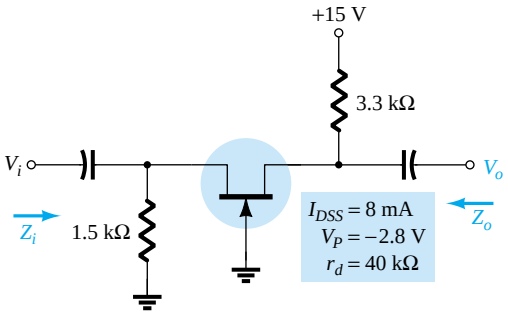


Figure 9.60 Problems 30, 31, and 49

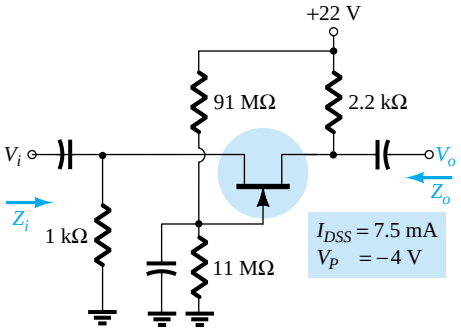


Figure 9.61 Problem 32

§ 9.8 Depletion-Type MOSFETs

33. Determine V_o for the network of Fig. 9.62 if $y_{os} = 20\ \mu S$.

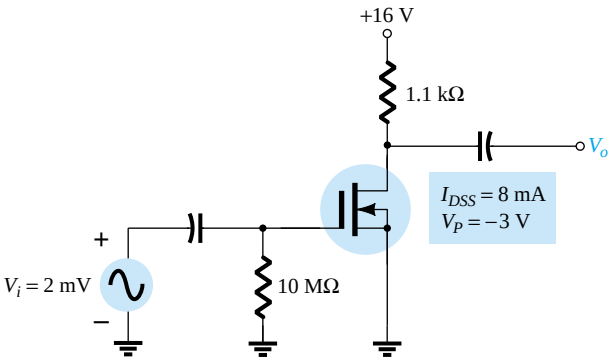


Figure 9.62 Problem 33

34. Determine Z_i, Z_o, and A_v for the network of Fig. 9.63 if $r_d = 60\ k\Omega$.

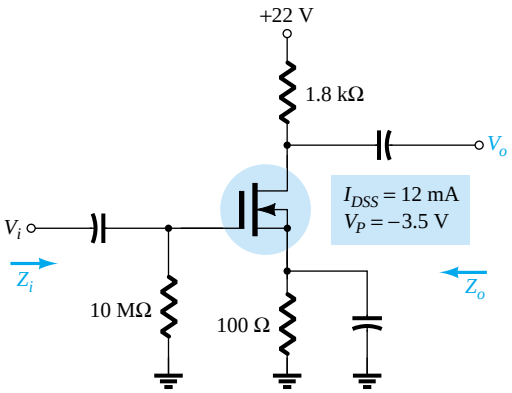


Figure 9.63 Problems 34, 35, and 50

35. Repeat Problem 34 if $r_d = 25\ \text{k}\Omega$.

36. Determine V_o for the network of Fig. 9.64 if $V_i = 4$ mV.

37. Determine Z_i, Z_o, and A_v for the network of Fig. 9.65.

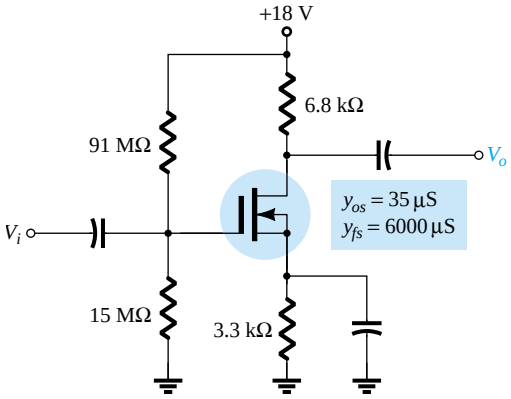


Figure 9.64 Problem 36

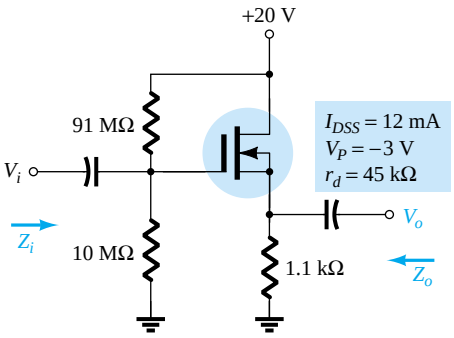


Figure 9.65 Problem 37

§ 9.10 E-MOSFET Drain-Feedback Configuration

38. Determine g_m for a MOSFET if $V_{GS(\text{Th})} = 3$ V and it is biased at $V_{GS_Q} = 8$ V. Assume $k = 0.3 \times 10^{-3}$.

39. Determine Z_i, Z_o, and A_v for the amplifier of Fig. 9.66 if $k = 0.3 \times 10^{-3}$.

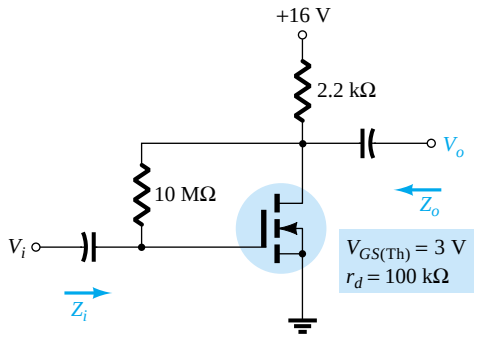


Figure 9.66 Problems 39, 40, and 51

40. Repeat Problem 39 if k drops to 0.2×10^{-3}. Compare results.

41. Determine V_o for the network of Fig. 9.67 if $V_i = 20$ mV.

42. Determine V_o for the network of Fig. 9.67 if $V_i = 4$ mV, $V_{GS(\text{Th})} = 4$ V, and $I_{D(\text{on})} = 4$ mA, with $V_{GS(\text{on})} = 7$ V and $y_{os} = 20\ \mu$S.

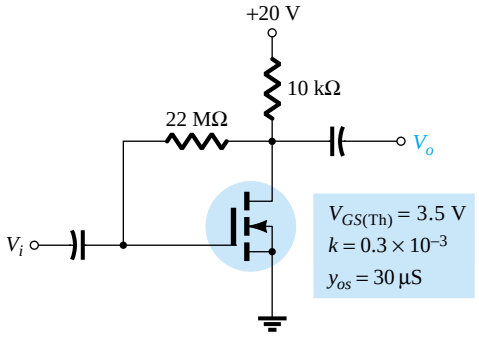


Figure 9.67 Problems 41 and 42

§ 9.11 E-MOSFET Voltage-Divider Configuration

43. Determine the output voltage for the network of Fig. 9.68 if $V_i = 0.8$ mV and $r_d = 40$ kΩ.

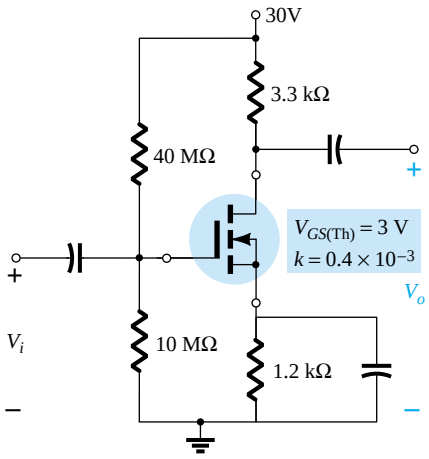


Figure 9.68 Problem 43

§ 9.12 Designing FET Amplifier Networks

44. Design the fixed-bias network of Fig. 9.69 to have a gain of 8.

45. Design the self-bias network of Fig. 9.70 to have a gain of 10. The device should be biased at $V_{GS_Q} = \frac{1}{3}V_P$.

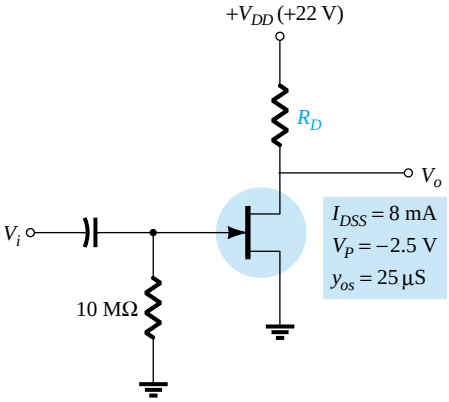


Figure 9.69 Problem 44

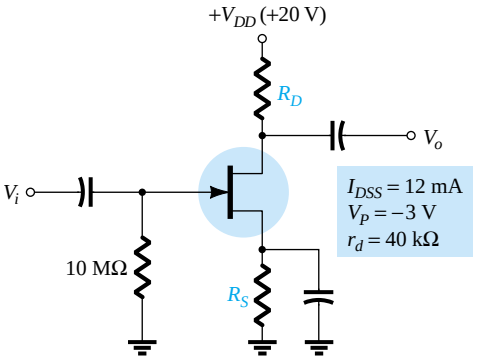


Figure 9.70 Problem 45

§ 9.13 PSpice Windows

46. Using PSpice Windows, determine the voltage gain for the network of Fig. 9.55.

47. Using PSpice Windows, determine the voltage gain for the network of Fig. 9.56.

48. Using PSpice Windows, determine the voltage gain for the network of Fig. 9.57.

49. Using PSpice Windows, determine the voltage gain for the network of Fig. 9.60.

50. Using PSpice Windows, determine the voltage gain for the network of Fig. 9.63.

51. Using PSpice Windows, determine the voltage gain for the network of Fig. 9.66.